

Chebyshev Greedy Algorithm for Nonlinear PDEs

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Abstract

We study the Chebyshev greedy algorithm (CGA) for convex nonlinear partial differential equation energies by shallow ReLU^k ridge functions, selected from finite candidate pools with inexact coefficient solves. A one-step estimate separates four contributions: coverage of the continuous dictionary by the pool, evaluation of the selection score, radial stationarity in the active space, and the accuracy of the coefficient optimization. For the nested spaces generated by the algorithm, an approximate-source projection bound keeps both the source remainder and the observed orthogonal increments. For the one-dimensional $p = 4$ energies, an exact expansion connects this bound to nonlinear energy errors. A separate power-type decrease hypothesis gives an explicit projection bound on a finite index range. For standard p -Laplacian energies with $p \geq 2$, an energy exponent α yields natural-distance and Sobolev exponents $\alpha/2$ and α/p . Eight one- and two-dimensional problems illustrate the estimates, and five of them compare finite-pool CGA with finite elements and with random features at equal numbers of independent coefficients. In the two $p = 4$ models the calibration-range bounds cover the computed errors on $N = 8$ – 32 but carry conservative constants. Continuing the same prefixes from $N = 32$ to $N = 64$ with those constants held fixed, the normalized-innovation sufficient condition holds on 27 and 28 of the 32 continuation transitions, respectively; this is stepwise numerical evidence and does not extend the calibration-range envelope. These results are finite-range observations; they establish neither an unconditional asymptotic rate nor a transfer from the finite pool to the continuous dictionary.

Keywords. Chebyshev greedy algorithm; nonlinear partial differential equation; shallow neural network; p -Laplacian; natural distance; finite dictionary; orthogonal greedy approximation.

Code and data. The source files, numerical code, raw inputs, and supplementary records are archived in the companion repository <https://github.com/RonzolYu/cga4PDE>. The repository also contains the configuration index, generated tables, figures, and the integrity information associated with the reported results. The supplementary files are `supplement/S1_experiment_index.md`, `supplement/S2_pool_generation.md`, `supplement/S2_window_certificate.md`, `supplement/S3_failures_and_statistics.md`, and `supplement/S4_reproduction.md`; the case index and figure map are `manifest/experiment_index_v2.csv` and `manifest/figure_table_map.csv`, and file integrity is recorded in `manifest/SHA256SUMS`.

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1 Introduction

Variational formulations turn a broad class of elliptic boundary-value problems into convex energy minimization. Classical finite elements approximate the admissible space by locally supported polynomials, and the relation between polynomial degree, mesh scale, and degrees of freedom is well understood (Brenner and Scott, 2008). Neural and ridge-function methods instead construct global nonlinear trial spaces, so that the approximation problem concerns the selection and reoptimization of atoms rather than the design of a mesh. The deep Ritz method illustrates the reach of variational neural solvers (E and Yu, 2018). The present work analyzes a deterministic greedy selection rule for such nonlinear elliptic energies.

Orthogonal greedy approximation (OGA) has a sharp Hilbert-space theory. Under suitable source and entropy conditions it produces optimal approximation rates (Siegel and Xu, 2022; Li and Siegel, 2024). Chebyshev-type greedy methods extend the basic descent argument to convex energies in Banach spaces (Temlyakov, 2013, 2014), and biorthogonal or inexact variants show how selection and correction errors enter the energy recursion (Dereventsov and Temlyakov, 2022). These results supply the approximation theory on which the analysis below relies. Applying them to a nonlinear partial differential equation (PDE) whose atoms are drawn from a finite candidate pool also requires accounting for numerical quadrature, finite dictionary approximation, and inexact coefficient minimization. The three effects are distinct: quadrature perturbs the evaluated score, a finite pool replaces the continuous supremum by a maximum over the pool, and an active-space solver returns only an approximate minimizer.

The dictionary used below consists of shallow ReLU^k ridge functions. Here $\text{ReLU}^k(s) = [s]_+^k$, with $[s]_+ = \max\{s, 0\}$, and the domain is a bounded subset of $\mathbb{R}^d$. Greedy neural training has been used successfully for PDEs (Siegel et al., 2023). A pool that is drawn once and then held fixed during the greedy solve provides one route to weak selection, by which we mean selection of an atom whose score is a prescribed fraction of the best score available in the full dictionary: Xu and Xu (2025, Theorem 3.1) prove a convex-energy bound under Hilbert smoothness and source/entropy assumptions, while their Theorems 5.1 and 5.2 quantify dictionary sizes that realize weak selection. Their analysis assumes exact coefficient minimization over the active span. The finite-pool inequality below separates the correction error from the score and coverage errors. The universality of such constructions for nonlinear convex variational problems is examined by Berná and Falcó (2025). Variation spaces and sharp entropy estimates for shallow networks provide the source and width information used in the conditional estimates (Siegel and Xu, 2023, 2024; Mao et al., 2026a).

The main contribution is a framework for analysing a nonlinear convergence rate on a concrete semilinear problem family. Section 5.3 identifies the Hessian comparison, invariant-neighbourhood, and same-dictionary entropy conditions under which the conditional rate follows. The numerical results provide finite-range evidence for the quantities entering these conditions; they do not by themselves establish an unconditional asymptotic rate.

The contribution has three parts. For the concrete semilinear family, we derive a local Hessian comparison and state the relative-score, invariant-neighbourhood, and entropy assumptions needed for the transfer argument. For the inexact method, we separate pool coverage, score quadrature, radial stationarity, and correction errors. The entropy statement uses the same projected, unrenormalized restarted dictionary and one fixed convention for dyadic entropy numbers.

The convex Chebyshev algorithms of Temlyakov (2013, 2014) and the inexact weak-greedy framework of Dereventsov and Temlyakov (2022) are closest to the present setting. The former supplies the generic descent argument, and the latter shows how approximate selection and correction enter an energy recurrence. The Hilbert OGA results of Siegel and Xu (2022); Li and Siegel

(2024) apply after a local transfer to a fixed Hessian metric; they do not by themselves control a nonlinear PDE score or selection from a finite pool. Xu and Xu (2025) quantify deterministic and randomized finite dictionaries under weak selection, but they use exact active-space minimization and do not analyze a degenerate p -energy. The present work examines the additional conditions needed for nonlinear scores, finite pools, and inexact active-space minimization.

For p -growth energies the metric that matches the energy is not the ambient Sobolev norm. The nonlinear map

$$V(\xi) = |\xi|^{(p-2)/2} \xi$$

defines the natural distance, as in the finite-element analysis of the p -Laplacian (Barrett and Liu, 1993; Diening and Kreuzer, 2008; Lee and Park, 2025). A squared V -distance has the scale of an energy gap, so an energy bound of order $m^{-\alpha}$ gives a V -distance bound of order $m^{-\alpha/2}$ and, for $p \geq 2$, a $W^{1,p}$ -error bound of order $m^{-\alpha/p}$. This conversion has the same form for every p , although the energy exponent may depend on p . We prove the equivalence for the standard pure, regularized, and reaction p -energies and apply the corresponding conversion to each of them.

The atomic source condition of Assumption 2.2 and the directional smoothness estimate of Assumption 3.3 apply to convex energies and yield an $O(m^{-1})$ energy bound for the ideal continuous CGA. The same arguments produce an explicit one-step inequality for finite-pool CGA, in which four contributions appear separately: continuous-dictionary coverage, score quadrature, radial active-space stationarity, and coefficient optimization.

Two conditional refinements use additional residual or Hessian information. The residual-comparison estimate combines power-type convexity with a lower bound for the dictionary dual seminorm in terms of the full dual norm. A local Hessian transfer uses coercivity, a dictionary norm comparison, and a relative score condition to obtain a weak-OGA estimate in a frozen Hilbert metric. Neither set of assumptions implies the other, and both are stated as conditional estimates.

The finite trajectory produced by the algorithm can also be analyzed without requiring uniform Hessian coercivity. For a fixed finite symmetric dictionary and a fixed starting index m_0 , an explicit approximation of the source with coefficient budget B and remainder σ gives a projection estimate

$$r_{m_0+j}^2 \leq \sigma^2 + \frac{4B^2}{\mathcal{W}_j}, \quad \mathcal{W}_j = \sum_{m=m_0}^{m_0+j-1} \frac{\theta_m^2}{\ell_m^2}.$$

Here r_m is the norm of the projection error at state m , ℓ_m is the innovation norm of the selected candidate, and $0 \leq \theta_m \leq \theta_m$ records the selected direction against the best score available in the frozen dictionary. For the one-dimensional $p = 4$ energies, whose density is quartic in the gradient, an exact expansion connects this projection error to the energy gap. The estimate applies to the active spaces actually generated and carries a source remainder; it does not assume that the target lies in the finite dictionary. The stronger initial-value form of the bound is stated and evaluated numerically. A second estimate uses the normalized innovation $\gamma_m = \theta_m S_m / \ell_m$. If $\gamma_m^2 \geq c_0 r_m^{2+2/\alpha}$, inversion of the exact projection recurrence bounds $r_{m_0+j}^2$ by $(r_{m_0}^{-2/\alpha} + c_0 j / \alpha)^{-\alpha}$. The same quartic comparison then gives the corresponding energy and metric bounds. Each condition is evaluated from the retained projection drops; on a finite index range, positivity alone does not single out a unique exponent.

The numerical study evaluates the estimates together with the conditions on which they rest. Eight representative cases cover linear reaction–diffusion, cubic and hyperbolic-sine semilinear energies, and pure, regularized, and reaction p -energies. Five of them compare finite-pool CGA, a random-feature model (RFM), and finite elements (FEM) at equal numbers of independent coefficients. Random features are used in their fixed-feature sense (Rahimi and Recht, 2007); conditioning

of linearized neural trial spaces is discussed by Chen et al. (2022) and Mao et al. (2026b). The comparison concerns coefficient counts and uses a common evaluator. It does not attribute an unproved convergence theorem to the compared methods and does not compare wall-clock efficiency. P1 and P3 elements provide low- and high-order mesh-based references, and P2 results are retained in the supplementary material.

For the finite-trajectory estimates we use the one-dimensional pure and regularized $p = 4$ trajectories. The calibration range is $N = 8\text{--}32$, and the subsequent range uses the same stored prefixes from $N = 33$ to 64 with the calibration constants fixed in advance. The reported fractions record the number of steps satisfying the stated sufficient condition on each finite range. They provide numerical evidence on the displayed trajectory; they do not verify continuous- dictionary coverage or an entropy estimate uniformly over increasing ranges. The approximate source is constructed once at the starting index from the fixed candidate and auxiliary reference dictionary. The source remainder is small relative to the starting residual but is not negligible at the terminal scale; it therefore limits the resolution of the source upper bound without giving a lower bound on the actual algorithmic error. Neither calculation verifies the continuous- dictionary coverage and entropy assumptions of the Hessian-transfer result, so the measured orders and fitted innovation powers are reported as finite-range observations.

The remainder is organized as follows. Section 2 defines the variational setting, dictionaries, and algorithms. Section 3 proves the basic descent estimates, and Section 4 develops the energy families and natural-distance conversion for $p \geq 2$. Section 5 combines residual comparison, Hessian transfer, finite-trajectory bounds, and fractional innovation. Section 6 tests these estimates on the computed trajectories; Section 7 presents equal-DOF comparisons, and Section 8 summarizes the conclusions and their scope.

2 Variational setting, dictionaries, and algorithms

Let X be a real reflexive Banach space. The abstract descent results apply to a proper convex functional $\mathcal{E} : X \rightarrow (-\infty, +\infty]$, with effective domain $\text{dom } \mathcal{E} = \{v \in X : \mathcal{E}(v) < +\infty\}$. Differentiability is assumed on an open neighborhood of the relevant iterates in the specified energy space, on which $\mathcal{E}$ is finite and continuously Fréchet differentiable. For an energy with a restricted effective domain, the additional topology and the neighborhood are stated explicitly. When an argument invokes a Hessian, its stated neighborhood is in addition contained in the twice differentiable part of the effective domain. Suppose that $\mathcal{E}$ has a unique minimizer u^* in the admissible space. The duality pairing between X^* and X is denoted by $\langle r, v \rangle$, and the energy defect is

$$\delta(u) = \mathcal{E}(u) - \mathcal{E}(u^*), \quad \delta_m = \delta(G_m).$$

The examples in Section 4.1 satisfy this setting after the Neumann nullspace has been removed where necessary.

2.1 Ridge dictionaries and admissibility

For an integer activation order $k \geq 1$, a raw ridge atom has the form

$$\varphi_{w,b}(x) = \text{ReLU}(w \cdot x + b)^k, \quad (w, b) \in \mathbb{R}^d \times \mathbb{R}.$$

An admissible dictionary consists of nonzero ridge functions satisfying the constraints of X , after the corresponding linear projection. We use a symmetric normalized dictionary $\mathcal{D}_X \subset X$:

$$g \in \mathcal{D}_X \implies -g \in \mathcal{D}_X, \quad \|g\|_X = 1 \quad (g \in \mathcal{D}_X).$$

When density is needed, we also require $\overline{\text{span } \mathcal{D}_X}^X = X$. We abbreviate $\mathcal{D}_X$ to $\mathcal{D}$ unless a different normalization is specified.

For pure-gradient Neumann energies the admissible space is

$$X_{p,\circ} = \left\{ v \in W^{1,p}(\Omega) : \int_{\Omega} v \, dx = 0 \right\}.$$

Each ridge is projected onto $X_{p,\circ}$ before the normalized dictionary is formed. Since an almost constant ridge may have arbitrarily small projected norm, normalization can amplify its derivatives. Hence uniform derivative bounds for the normalized dictionary require a positive lower bound on the projected norms. Zero projected atoms are excluded from the analytical dictionary; the corresponding nondegeneracy assumption is stated whenever it is needed.

Definition 2.1 (Atomic hull and atomic norm). For a symmetric normalized dictionary $\mathcal{D} \subset X$, define

$$K_1(\mathcal{D}) = \overline{\left\{ \sum_{j=1}^J c_j g_j : J < \infty, g_j \in \mathcal{D}, \sum_{j=1}^J |c_j| \leq 1 \right\}}^X,$$

and

$$\|v\|_{\mathcal{A}(\mathcal{D})} = \inf\{B > 0 : v/B \in K_1(\mathcal{D})\}.$$

For a bounded set $K \subset X$, its n -th dyadic entropy number is

$$\varepsilon_n(K; X) = \inf \left\{ h > 0 : \exists x_1, \dots, x_{2^{n-1}} \in X, K \subset \bigcup_{j=1}^{2^{n-1}} (x_j + hB_X) \right\}, \quad n \geq 1,$$

where $B_X = \{x \in X : \|x\|_X \leq 1\}$. Thus the covering balls may have centers in X . This 2^{n-1} -ball convention is used throughout the manuscript. We write $A \lesssim B$ when $A \leq CB$ with a constant independent of the iteration index; a subscript on $\lesssim$ lists the permitted parameter dependence, and $A \simeq B$ denotes the two corresponding inequalities.

Assumption 2.2 (Atomic source). There is a finite $B > 0$, with $B \geq \|u^*\|_{\mathcal{A}(\mathcal{D})}$, such that $u^*/B \in K_1(\mathcal{D})$.

This is a property of the target relative to the *theoretically normalized* dictionary. It is not interchangeable with the coefficient ℓ^1 -norm produced by a discretely normalized algorithm. In one space dimension, Proposition 2.3 gives a sufficient condition for finite atomic norm.

Proposition 2.3 (A one-dimensional ReLU^k source class). *Let $I = (0, 1)$, $k \geq 1$, $1 \leq p < \infty$, and $X = W^{1,p}(I)$. Suppose that the continuous dictionary contains the normalized nonzero truncated powers $(x - t)_+^k / \|(\cdot - t)_+^k\|_X$, $0 \leq t < 1$, together with normalized fully active ridge functions spanning $\mathbb{P}_k$. If $v \in W^{k+1,1}(I)$, then v belongs to the atomic space and*

$$\|v\|_{\mathcal{A}(\mathcal{D})} \leq C_{k,p} \left(\sum_{j=0}^k \left| v^{(j)}(0) \right| + \left\| v^{(k+1)} \right\|_{L^1(I)} \right),$$

where $C_{k,p}$ also accounts for the fixed exponent p , the interval, and the normalization of the raw atoms. It is a continuous-dictionary source constant and is not the coefficient ℓ^1 -norm returned by a discretely normalized numerical solve.

The proof, given in Section A, uses the integral remainder formula. Multivariate source and entropy inputs are instead supplied by shallow-network variation-space and width theorems (Siegel and Xu, 2023, 2024; Mao et al., 2026a); their domain and norm hypotheses must be checked separately for each application.

Lemma 2.4 (Projection and renormalization). *Let $P : X \rightarrow X_0$ be bounded and linear. If a subdictionary $\mathcal{D}_0 \subset \mathcal{D}$ satisfies*

$$0 < c_0 \leq \|Pg\|_{X_0} \leq C_0 < \infty \quad (g \in \mathcal{D}_0),$$

and $\tilde{\mathcal{D}}_0 = \{Pg / \|Pg\|_{X_0} : g \in \mathcal{D}_0\}$, then

$$PK_1(\mathcal{D}_0) \subset C_0 K_1(\tilde{\mathcal{D}}_0).$$

The same computation also shows that a bounded linear map increases every entropy radius by at most $\|P\|$, while the inverse normalization loses at most the factor c_0^{-1} :

$$K_1(\tilde{\mathcal{D}}_0) \subset c_0^{-1} PK_1(\mathcal{D}_0).$$

Proof. For an absolute convex combination, write $Pg = \|Pg\|_{X_0} \tilde{g}$ term by term and bound the new coefficient sum by C_0 . Pass to the closure for the first claim. Applying P to an h -cover gives a $\|P\|$ h -cover. Conversely, $Pg / \|Pg\|_{X_0} = c_0^{-1} P[(c_0 / \|Pg\|_{X_0})g]$, and the coefficient $c_0 / \|Pg\|_{X_0}$ is at most one. Taking absolute convex combinations and closures proves the reverse inclusion. This statement concerns convex hulls; it does not assert that pointwise renormalization is Lipschitz near null projected atoms. The image of $K_1(\mathcal{D}_0)$ is closed because this bounded closed convex subset of the reflexive space X is weakly compact, and its image under P is weakly compact in X_0 . $\square$

Corollary 2.5 (Transfer of source and entropy inputs). *In the setting of Lemma 2.4, suppose $u^* = Pu^\dagger$, $u^\dagger/B \in K_1(\mathcal{D}_0)$, and*

$$\varepsilon_n(K_1(\mathcal{D}_0); X) \leq C_{\text{ent}} n^{-\beta}.$$

Then

$$\frac{u^*}{BC_0} \in K_1(\tilde{\mathcal{D}}_0), \quad \varepsilon_n(K_1(\tilde{\mathcal{D}}_0); X_0) \leq c_0^{-1} \|P\| C_{\text{ent}} n^{-\beta}.$$

Thus a ReLU source/convex-hull entropy theorem on an extended parameter domain passes to the projected, renormalized feasible dictionary whenever the nondegeneracy bounds in Lemma 2.4 hold.

Proof. The first assertion follows from $PK_1(\mathcal{D}_0) \subset C_0 K_1(\tilde{\mathcal{D}}_0)$. For the second, use $K_1(\tilde{\mathcal{D}}_0) \subset c_0^{-1} PK_1(\mathcal{D}_0)$ and map an entropy cover through P . $\square$

2.2 Dictionary residuals

For $r \in X^*$, introduce the dictionary dual seminorm

$$\|r\|_{\mathcal{D}^*} = \sup_{g \in \mathcal{D}_X} |\langle r, g \rangle|.$$

This quantity measures the derivative in the available dictionary directions. It is a seminorm; if the dictionary has dense linear span, its vanishing implies that the derivative is zero.

2.3 Ideal and finite-pool Chebyshev greedy algorithms

The ideal CGA combines weak selection with exact minimization over the enlarged space.

Algorithm 2.6 (Continuous weak Chebyshev greedy algorithm). Assume $\mathcal{E}(0) < \infty$, choose weakness numbers $t_m \in (0, 1]$, and set $G_0 = 0$, $\Phi_0 = \{0\}$. Given G_m and $\Phi_m = \text{span}\{g_1, \dots, g_m\}$:

- (i) if $\|\mathcal{E}'(G_m)\|_{\mathcal{D}^*} = 0$, stop; otherwise choose $g_{m+1} \in \mathcal{D}_X$ with descent sign such that

$$-\langle \mathcal{E}'(G_m), g_{m+1} \rangle \geq t_m \|\mathcal{E}'(G_m)\|_{\mathcal{D}^*};$$

- (ii) set $\Phi_{m+1} = \text{span}(\Phi_m, g_{m+1})$ and compute

$$G_{m+1} = \arg \min_{v \in \Phi_{m+1}} \mathcal{E}(v).$$

For a general continuous dictionary the supremum need not be attained. Thus $t_m < 1$ is understood as approximate-supremum selection. The endpoint $t_m = 1$ is used only when attainment has been proved, for example on a compact nondegenerate parameter set with continuous score. The finite-pool method stops, with a recorded reason, if the unused pool is empty, every remaining score is below the stationarity threshold, every candidate fails the rank or acceptance test, or the attempt budget is reached. In an unconstrained active-space solve, changing an atom's sign leaves the span unchanged; the descent sign required by the proof is therefore not a further quantity that the implementation has to resolve.

Exact reoptimization yields the variational orthogonality

$$\langle \mathcal{E}'(G_m), v \rangle = 0 \quad (v \in \Phi_m). \quad (1)$$

The second algorithm describes the computed object. It uses a finite pool $\mathcal{P}_m$, an evaluated score $\tilde{s}_m$, and a correction computed to a finite tolerance.

Algorithm 2.7 (Finite-pool inexact CGA). Set $\hat{G}_0 = 0$, $\hat{\Phi}_0 = \{0\}$. At step m , form a finite admissible pool $\mathcal{P}_m \subset \mathcal{D}$. Let

$$s_m(g) = \left| \langle \mathcal{E}'(\hat{G}_m), g \rangle \right|, \quad \tilde{s}_m(g) \geq 0.$$

Choose $t_m^{\text{pool}} \in (0, 1]$ and $\hat{g}_{m+1} \in \mathcal{P}_m$ satisfying

$$\tilde{s}_m(\hat{g}_{m+1}) \geq t_m^{\text{pool}} \max_{g \in \mathcal{P}_m} \tilde{s}_m(g),$$

give it the true descent sign, and define $\hat{\Phi}_{m+1} = \text{span}(\hat{\Phi}_m, \hat{g}_{m+1})$. Compute a coefficient vector whose state obeys

$$\mathcal{E}(\hat{G}_{m+1}) \leq \inf_{v \in \hat{\Phi}_{m+1}} \mathcal{E}(v) + \varepsilon_m^{\text{opt}}, \quad \varepsilon_m^{\text{opt}} \geq 0.$$

2.4 Finite-pool error quantities

The continuous-dictionary supremum, the exact maximum over the finite pool, and the numerically evaluated maximum are distinct quantities. We impose the following score-error and pool-coverage conditions on any step for which a positive coverage bound is used; finiteness of the pool alone does not imply a uniform positive lower bound.

$$\sup_{g \in \mathcal{P}_m} |\tilde{s}_m(g) - s_m(g)| \leq \xi_m, \quad (2)$$

$$\max_{g \in \mathcal{P}_m} s_m(g) \geq \tau_m \sup_{g \in \mathcal{D}} s_m(g), \quad 0 < \tau_m \leq 1. \quad (3)$$

Lemma 2.8 (True score selected from an evaluated pool). *Under Equations (2) and (3),*

$$s_m(\hat{g}_{m+1}) \geq t_m^{\text{pool}} \tau_m \left\| \mathcal{E}'(\hat{G}_m) \right\|_{\mathcal{D}^*} - (1 + t_m^{\text{pool}}) \xi_m. \quad (4)$$

Proof. Insert the score error before and after the finite-pool maximization:

$$\begin{aligned} s_m(\hat{g}_{m+1}) &\geq \tilde{s}_m(\hat{g}_{m+1}) - \xi_m \\ &\geq t_m^{\text{pool}} \max_{g \in \mathcal{P}_m} \tilde{s}_m(g) - \xi_m \\ &\geq t_m^{\text{pool}} \max_{g \in \mathcal{P}_m} s_m(g) - (1 + t_m^{\text{pool}}) \xi_m, \end{aligned}$$

and use Equation (3). $\square$

If the implemented correction minimizes a discrete training objective $\mathcal{E}_h$, a single terminal comparison between $\mathcal{E}_h$ and the analysis energy $\mathcal{E}$ is not a uniform error bound. The appropriate deterministic statement is the following.

Let $G_{m+1}^\sharp$ denote an exact minimizer of $\mathcal{E}$ over $\hat{\Phi}_{m+1}$. For an energy level Λ_m^E chosen so that both $G_{m+1}^\sharp$ and the computed state $\hat{G}_{m+1}$ are included, define the relevant sublevel set and its active-space restriction by

$$\mathcal{L}_m = \{v \in X : \mathcal{E}(v) \leq \Lambda_m^E\}, \quad \mathcal{L}_m^{\text{act}} = \hat{\Phi}_{m+1} \cap \mathcal{L}_m. \quad (5)$$

Thus $\mathcal{L}_m^{\text{act}}$ is the part of the current finite-dimensional trial space on which the training and analysis energies must be compared; in particular, $\inf_{v \in \mathcal{L}_m^{\text{act}}} \mathcal{E}(v) = \mathcal{E}(G_{m+1}^\sharp)$.

Lemma 2.9 (Transfer from the discrete energy to the analysis energy). *Suppose on the active-space sublevel set $\mathcal{L}_m^{\text{act}}$ in Equation (5) that*

$$\sup_{v \in \mathcal{L}_m^{\text{act}}} |\mathcal{E}_h(v) - \mathcal{E}(v)| \leq \zeta_m^E.$$

If the computed state satisfies

$$\mathcal{E}_h(\hat{G}_{m+1}) \leq \inf_{v \in \mathcal{L}_m^{\text{act}}} \mathcal{E}_h(v) + \varepsilon_m^{\text{opt,h}},$$

then Algorithm 2.7 holds with $\varepsilon_m^{\text{opt}} \leq \varepsilon_m^{\text{opt,h}} + 2\zeta_m^E$.

Proof. Apply the uniform energy difference once at $\hat{G}_{m+1}$ and once at $G_{m+1}^\sharp$. Both states belong to $\mathcal{L}_m^{\text{act}}$, and the latter minimizes the analysis energy over the full active space. $\square$

3 Atomic descent and finite-pool inexactness

This section converts the distinctions of Section 2 into energy inequalities. The standard atomic residual estimate is derived first, since its orthogonality hypothesis is weakened in the finite-tolerance algorithm.

Lemma 3.1 (Atomic residual bound). *Under Assumption 2.2 and the exact orthogonality Equation (1),*

$$\left\| \mathcal{E}'(G_m) \right\|_{\mathcal{D}^*} \geq \frac{\delta_m}{B}. \quad (6)$$

Proof. Convexity and $G_m \in \Phi_m$ give

$$\delta_m \leq \langle \mathcal{E}'(G_m), G_m - u^* \rangle = -\langle \mathcal{E}'(G_m), u^* \rangle.$$

Approximate $u^*/B \in K_1(\mathcal{D})$ by absolute convex combinations of atoms. The pairing of each combination with $\mathcal{E}'(G_m)$ is bounded by the dictionary dual seminorm, and passage to the limit gives Equation (6). $\square$

Definition 3.2 (p -growth density). Let $p \geq 2$, write $z = (s, \xi) \in \mathbb{R} \times \mathbb{R}^d$, and omit the scalar component s for a pure-gradient energy. A twice differentiable convex density $\Psi(x, z)$ has p -growth on the relevant range if, for almost every x ,

$$\begin{aligned} c_0 |z|^p - c_1 &\leq \Psi(x, z) \leq c_2(1 + |z|^p), \\ \|D_z^2 \Psi(x, z)\| &\leq c_3(1 + |z|)^{p-2}, \end{aligned}$$

with positive constants independent of x and z in that range. The first line controls bounded energy sublevels, whereas the Hessian bound is the part used in the directional estimate below.

Assumption 3.3 (Directional quadratic upper bound). There is $L > 0$ such that, on the trajectory sublevel set, for every $g \in \mathcal{D}$ and $|\lambda| \leq 1$,

$$\mathcal{E}(u + \lambda g) \leq \mathcal{E}(u) + \lambda \langle \mathcal{E}'(u), g \rangle + \frac{L}{2} \lambda^2. \quad (7)$$

For $a \geq 0$, define the maximal decrease supplied by the directional quadratic bound by

$$H_L(a) := \max_{0 \leq \lambda \leq 1} \left(\lambda a - \frac{L}{2} \lambda^2 \right) = \begin{cases} a^2/(2L), & 0 \leq a \leq L, \\ a - L/2, & a > L. \end{cases}$$

The assumption is directional; it does not require a globally Lipschitz derivative on the full Banach space. For the standard p -energies, bounded $W^{1,p}$ sublevels and an X -normalized dictionary already give the needed estimate by Hölder's inequality; an L^∞ atom bound is only a convenient stronger condition for more general densities. Indeed, for $F(\lambda) = \mathcal{E}(u + \lambda g)$, the diffusion part of F'' is controlled by

$$C_p \int_{\Omega} (|\nabla u| + |\lambda| |\nabla g| + \varepsilon)^{p-2} |\nabla g|^2 \, dx,$$

and Hölder's inequality gives $F''(\lambda) \leq C_p (\|\nabla u\|_{L^p} + \|\nabla g\|_{L^p} + \varepsilon |\Omega|^{1/p})^{p-2} \|\nabla g\|_{L^p}^2$ for $|\lambda| \leq 1$. A reaction term is handled similarly. Removing almost-null projected atoms remains important for the numerical parameterization and for uniform source/entropy constants, but it is not necessary for this abstract L^p estimate once the atoms are already normalized in the energy space.

Theorem 3.4 (Generic energy baseline). *Suppose Assumptions 2.2 and 3.3 hold and $t_m \geq t > 0$. Then the exact continuous CGA satisfies*

$$\delta_m \leq \frac{C(B, L, t, \delta_0)}{m+1}. \quad (8)$$

Proof. The selected descent score is at least $a_m = t\delta_m/B$ by Lemma 3.1. Test the new active space with $G_m + \lambda g_{m+1}$ and choose $\lambda_m = \min\{1, a_m/L\}$. If $a_m \leq L$, then

$$\delta_{m+1} \leq \delta_m - \frac{t^2}{2LB^2} \delta_m^2. \quad (9)$$

If $a_m > L$, the unit step reduces the energy by at least $a_m/2 > L/2$; such steps occur only finitely often because the energy is bounded below. After this finite prefix, taking reciprocals in Equation (9) and summing proves Equation (8). The truncation at one is essential because Assumption 3.3 is stated only on unit dictionary segments. $\square$

The standard p -energy setting used below is

$$\mathcal{E}_p(u) = \int_{\Omega} \frac{1}{p} |\nabla u|^p dx - \ell(u), \quad \mathcal{E}_{p,\varepsilon}(u) = \int_{\Omega} \frac{(\varepsilon^2 + |\nabla u|^2)^{p/2} - \varepsilon^p}{p} dx - \ell(u),$$

with the zero-mean space in the pure and regularized Neumann cases. With a reaction term we use

$$\mathcal{E}_{p,R}(u) = \int_{\Omega} \frac{|\nabla u|^p + |u|^p}{p} dx - \ell(u)$$

on $W^{1,p}(\Omega)$. The complete assumptions and spaces are specified in Section 4.

3.1 Inexact selection and correction

Write $\hat{\delta}_m = \mathcal{E}(\hat{G}_m) - \mathcal{E}(u^*)$. In place of exact orthogonality, suppose that

$$\left| \left\langle \mathcal{E}'(\hat{G}_m), \hat{G}_m \right\rangle \right| \leq \eta_m^{\text{stat}}. \quad (10)$$

The decomposition contains one multiplicative resolution factor and three additive error contributions: $t_m^{\text{pool}} \tau_m$ records weak selection and continuous-dictionary coverage, ξ_m records score evaluation, η_m^{stat} is the radial active-space stationarity defect of Equation (10) alone, and $\varepsilon_m^{\text{opt}}$ records the energy accuracy of the correction. The full active-space dual Karush–Kuhn–Tucker (KKT) residual is denoted separately by r_m^{KKT} below. In the terminology of the greedy-with-errors framework of Dereventsov and Temlyakov (2022, Section 3), $t_m^{\text{pool}} \tau_m$ is the effective weakness parameter and $\varepsilon_m^{\text{opt}}$ is an energy-evaluation/correction error. The quantities ξ_m and η_m^{stat} separate the finite-score and biorthogonality contributions that the abstract framework combines into a single stepwise inexactness allowance.

Theorem 3.5 (Finite-pool one-step descent). *Assume Assumptions 2.2 and 3.3, Equations (2) and (3), and Equation (10) on the numerical trajectory. Define*

$$a_m = \left[t_m^{\text{pool}} \tau_m \frac{\hat{\delta}_m - \eta_m^{\text{stat}}}{B} - (1 + t_m^{\text{pool}}) \xi_m \right]_+.$$

Here $[x]_+ = \max\{x, 0\}$. Then

$$\hat{\delta}_{m+1} \leq \hat{\delta}_m - H_L(a_m) + \varepsilon_m^{\text{opt}}. \quad (11)$$

If, for a fixed absorption constant $\vartheta \in (0, 1)$,

$$\eta_m^{\text{stat}} + \frac{B(1 + t_m^{\text{pool}})}{t_m^{\text{pool}} \tau_m} \xi_m \leq \vartheta \hat{\delta}_m, \quad (12)$$

the effective weakness $\inf_{m \geq m_0} t_m^{\text{pool}} \tau_m$ is positive, and there exist a finite index m_0 and $0 \leq \zeta < 1$ such that

$$\varepsilon_m^{\text{opt}} \leq \zeta H_L(a_m), \quad m \geq m_0,$$

then $\hat{\delta}_m = O((m - m_0 + 1)^{-1})$ for the tail. The finite prefix $m < m_0$ is not included in this rate statement.

Proof. Convexity and Equation (10) give

$$\widehat{\delta}_m \leq \left\langle \mathcal{E}'(\widehat{G}_m), \widehat{G}_m - u^* \right\rangle \leq \eta_m^{\text{stat}} + B \left\| \mathcal{E}'(\widehat{G}_m) \right\|_{\mathcal{D}^*}.$$

Combining this estimate with Lemma 2.8 bounds the true score of the selected atom below by a_m . The inexact correction and the directional quadratic upper bound imply, for $0 \leq \lambda \leq 1$,

$$\widehat{\delta}_{m+1} \leq \widehat{\delta}_m - \lambda a_m + \frac{L}{2} \lambda^2 + \varepsilon_m^{\text{opt}}.$$

Set $\lambda = \min\{1, a_m/L\}$ to obtain Equation (11). Under Equation (12), a_m is a fixed positive fraction of $\widehat{\delta}_m$. For $m \geq m_0$, the displayed ζ -condition absorbs the correction error into the principal decrease, leaving the recurrence used in Theorem 3.4 with modified constants. $\square$

Remark 3.6 (Error saturation). Equation (11) shows how error saturation arises. A pool whose τ_m decreases, a score error ξ_m comparable with the best resolvable score, or a correction error $\varepsilon_m^{\text{opt}}$ comparable with the predicted decrease can each stop the recursion. The equation does not identify which defect dominates from a plotted error curve alone.

The optimization error can be related to a KKT residual only after a norm and a strong-convexity constant have been fixed. We say that $\mathcal{E}|_{\Phi}$ is μ_{Φ} -strongly convex in the norm $\|\cdot\|_{\Phi}$ if, for all $u, v \in \Phi$,

$$\mathcal{E}(v) \geq \mathcal{E}(u) + \langle \mathcal{E}'(u), v - u \rangle + \frac{\mu_{\Phi}}{2} \|v - u\|_{\Phi}^2.$$

Here μ_{Φ} is a constant on the finite-dimensional active space Φ ; it cannot be extrapolated to a strong-convexity constant on the full space without an additional argument.

Lemma 3.7 (KKT residual to correction error). *Let Φ be finite dimensional and let $\mathcal{E}|_{\Phi}$ be μ_{Φ} -strongly convex in a norm $\|\cdot\|_{\Phi}$. If $G_{\Phi}^{\sharp}$ is the exact minimizer and*

$$r^{\text{KKT}} = \left\| \mathcal{E}'(\widehat{G})|_{\Phi} \right\|_{\Phi^*},$$

then

$$0 \leq \mathcal{E}(\widehat{G}) - \mathcal{E}(G_{\Phi}^{\sharp}) \leq \frac{(r^{\text{KKT}})^2}{2\mu_{\Phi}}. \quad (13)$$

The same residual also satisfies

$$\left| \left\langle \mathcal{E}'(\widehat{G}), \widehat{G} \right\rangle \right| \leq r^{\text{KKT}} \left\| \widehat{G} \right\|_{\Phi}. \quad (14)$$

Proof. Strong convexity at $\widehat{G}$ gives

$$\mathcal{E}(G_{\Phi}^{\sharp}) \geq \mathcal{E}(\widehat{G}) + \left\langle \mathcal{E}'(\widehat{G}), G_{\Phi}^{\sharp} - \widehat{G} \right\rangle + \frac{\mu_{\Phi}}{2} \left\| G_{\Phi}^{\sharp} - \widehat{G} \right\|_{\Phi}^2.$$

Apply the dual inequality and optimize the resulting quadratic in the distance to obtain Equation (13); Equation (14) is the dual inequality with $v = \widehat{G}$. $\square$

4 Natural geometry for variational energies

Let $\Omega \subset \mathbb{R}^d$ be bounded, connected, and Lipschitz. We consider diffusion and reaction equations with natural Neumann boundary conditions. The regularization parameter $\varepsilon \geq 0$ is fixed; the regularized model uses $\varepsilon > 0$. The linear and semilinear equations are

$$-\Delta u + u = f, \quad -\Delta u + u^3 = f, \quad -\Delta u + \sinh u = f.$$

The standard p -structure models are

$$-\operatorname{div} A_p(\nabla u) = f, \quad -\operatorname{div} A_{p,\varepsilon}(\nabla u) = f, \quad -\operatorname{div} A_p(\nabla u) + |u|^{p-2}u = f,$$

where $A_p(\xi) = |\xi|^{p-2}\xi$, $A_p(0) = 0$, and $A_{p,\varepsilon}(\xi) = (\varepsilon^2 + |\xi|^2)^{(p-2)/2}\xi$. The boundary flux is zero. For the pure and regularized Neumann problems, the load annihilates constants and the zero-mean representative is used. The equations are understood weakly when the load is not a regular function.

4.1 Variational energies and feasible spaces

The three p -energies are defined in Equations (18) to (20). The remaining models have the following energies, written for $d \leq 2$:

$$\mathcal{E}_{\text{lin}}(u) = \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{2} u^2 \right) dx - \langle f, u \rangle, \quad (15)$$

$$\mathcal{E}_{\text{cub}}(u) = \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{4} u^4 \right) dx - \langle f, u \rangle, \quad (16)$$

$$\mathcal{E}_{\text{sinh}}(u) = \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \cosh u - 1 \right) dx - \langle f, u \rangle. \quad (17)$$

The pairing is the energy-space duality; for a regular load it is the corresponding integral. The additive -1 in Equation (17) normalizes the potential and does not change the Euler-Lagrange equation.

Assumption 4.1 (Data and feasible spaces for the six models). The following model-specific conditions are imposed.

- (i) For the linear and cubic models, $f \in (H^1(\Omega))^*$. For the sinh model, the convex integral functional has a load $f \in (H^1(\Omega))^*$, and the integral functional is allowed to take the value $+\infty$, its effective domain is nonempty, and local derivative/Hessian statements are restricted to an L^∞ -bounded neighborhood.
- (ii) For the pure and regularized p -energies, $f \in X_{p,\circ}^*$, or equivalently the Neumann load annihilates constants before restriction to $X_{p,\circ}$.
- (iii) For the reaction p -energy, $f \in (W^{1,p})^*$. In the regularized model, the parameter ε in Equation (19) is fixed and positive.

Proposition 4.2 (Well-posedness). *Under Assumption 4.1, Equations (15) to (17) have unique minimizers in $H^1(\Omega)$; Equations (18) and (19) have unique minimizers in $X_{p,\circ}$; and Equation (20) has a unique minimizer in $W^{1,p}(\Omega)$.*

Proof. The positive zero-order term in $\mathcal{E}_{\text{lin}}$ controls the complete H^1 -norm. For the cubic energy, the L^4 term controls constants, while its combination with the gradient and Young's inequality

gives coercivity. Strict convexity follows from the gradient term and $t \mapsto t^4$. For the sinh energy, $\cosh t - 1 \geq t^2/2$ gives coercivity. Along a weakly convergent H^1 sequence, compactness in L^2 , almost-everywhere convergence of a subsequence, and Fatou's lemma give lower semicontinuity of the potential.

For the pure and regularized p -energies, the zero-mean Poincaré inequality turns the gradient seminorm into a norm, and the diffusion densities are strictly convex. For the reaction model, the gradient and scalar p -terms control the full $W^{1,p}$ -norm. The direct method gives existence in each case, and strict convexity gives uniqueness. $\square$

The local Hessian structure of each model is needed for the transfer estimates of Section 5.3. The bilinear forms at the exact solution are

$$\begin{aligned} a_{\text{lin}}(v, w) &= \int_{\Omega} (\nabla v \cdot \nabla w + vw) \, dx, \\ a_{\text{cub},*}(v, w) &= \int_{\Omega} (\nabla v \cdot \nabla w + 3(u^*)^2 vw) \, dx, \\ a_{\text{sinh},*}(v, w) &= \int_{\Omega} (\nabla v \cdot \nabla w + \cosh(u^*) vw) \, dx. \end{aligned}$$

For a p -diffusion density, the Hessian matrix in gradient variables is

$$|\xi|^{p-2} I + (p-2) |\xi|^{p-4} \xi \otimes \xi$$

at nonzero ξ in the pure case, with $|\xi|^2$ replaced by $\varepsilon^2 + |\xi|^2$ in the regularized case. The local Hessian statements that follow concern $p \geq 2$.

Proposition 4.3 (Sufficient continuous local Hessian conditions). *In dimensions $d \leq 2$, the cubic Hessian is locally Lipschitz as a bilinear form on H^1 , and it is coercive at any solution $u^* \neq 0$. The sinh Hessian is coercive and is locally Lipschitz on an L^∞ -bounded neighborhood, with the state perturbation measured in H^1 . For fixed $\varepsilon > 0$, the regularized p -Hessian is uniformly elliptic, with lower bound proportional to ε^{p-2} ; its local Lipschitz continuity is valid in the bounded-gradient topology $W^{1,\infty}$, not in the H^1 topology asserted for the cubic model.*

Proof. For the cubic multiplier,

$$\left| \int_{\Omega} 3(u^2 - (u^*)^2) vw \, dx \right| \leq C \|u - u^*\|_{L^4} \|u + u^*\|_{L^4} \|v\|_{L^4} \|w\|_{L^4},$$

and $H^1 \hookrightarrow L^4$ for $d \leq 2$. If $u^* \neq 0$, coercivity follows by contradiction: a unit H^1 -sequence with vanishing quadratic form has gradients tending to zero and, by compact embedding, converges in L^4 to a constant; the weighted term then forces that constant to vanish. On $|u|, |u^*| \leq R_\infty$, the mean value theorem gives $|\cosh u - \cosh u^*| \leq \sinh(R_\infty) |u - u^*|$, after which the chosen local multiplication estimate applies. In the regularized model, the radial and tangential eigenvalues are comparable to $(\varepsilon^2 + |\xi|^2)^{(p-2)/2}$; the lower bound is therefore proportional to ε^{p-2} . The matrix-valued mean value theorem gives Hessian stability when the gradients and their perturbation are controlled in L^∞ . Such control is not a consequence of a bounded H^1 neighborhood. $\square$

Proposition 4.4 (Restricted stability for the regularized p -energy). *Let $Y_M \subset W^{1,\infty}(\Omega)$ be finite dimensional and let $\|\cdot\|_{0,M}$ be any norm on Y_M . For fixed $\varepsilon > 0$, bounded subsets of Y_M have finite restricted constants L_M on the corresponding finite-dimensional segments. If the reference norm is chosen so that the gradient seminorm is a norm on the feasible space, then $\mu_M > 0$. The resulting constants may depend on M , the inverse inequality on Y_M , and ε .*

Table 1: Hessian structure and the conclusion available for each experimental energy.

Model	Frozen Hessian	Scope of the conclusion
Linear	Constant H^1 inner product	CGA and energy-inner-product OGA coincide under exact correction, the same dictionary, and the same selection rule.
Cubic	Gradient form plus $3(u^*)^2$	Locally stable for $d \leq 2$ and coercive when $u^* \neq 0$; transfer still needs relative score resolution.
Sinh	Gradient form plus $\cosh(u^*)$	Coercive; Hessian Lipschitz is stated on an L^∞ -bounded neighborhood, and the numerical use is finite-dimensional.
Regularized p	Uniformly elliptic for fixed $\varepsilon > 0$	Continuous transfer needs bounded-gradient stability; finite-dimensional transfer has explicit M -dependent constants.
Pure p , $p > 2$	Weight can vanish at $\nabla u^* = 0$	Use global p -geometry or the explicit finite-range estimate.
Reaction p , $p > 2$	Gradient and scalar weights can both vanish	The reaction restores full-space energy coercivity, not uniform frozen-Hessian coercivity.

Proof. Norm equivalence on Y_M converts a $\|\cdot\|_{0,M}$ -bounded set into a $W^{1,\infty}$ -bounded set. The bounded-gradient Hessian estimate in the preceding proof then gives finite L_M . Uniform ellipticity for fixed ε , together with the assumed treatment of constant directions, gives $\mu_M > 0$. No constant uniform in M or $\varepsilon \downarrow 0$ follows. $\square$

Energy coercivity and frozen-Hessian coercivity are distinct. A positive reaction term does not imply that the frozen p -Hessian is uniformly positive at points where both the solution and its gradient vanish, and the continuous degeneracy of the pure model does not preclude the restricted coercivity in the corresponding finite-dimensional metric.

4.2 Standard p -structure and natural distance

For standard p -structure energies, we relate the CGA energy error to the natural distance associated with the diffusion operator. The argument uses the pointwise structure of the density in addition to its growth. The pointwise inequalities are standard; the step taken here is to integrate them into an energy-to-metric estimate for the three energy families below, together with the matching $W^{1,p}$ conversion. Let $\Omega \subset \mathbb{R}^d$ be bounded, connected, and Lipschitz, and let $p \geq 2$. On the zero-mean space $X_{p,0}$, define the pure and regularized energies

$$\mathcal{E}_p(u) = \int_{\Omega} \frac{1}{p} |\nabla u|^p \, dx - \langle f, u \rangle, \quad (18)$$

$$\mathcal{E}_{p,\varepsilon}(u) = \int_{\Omega} \frac{(\varepsilon^2 + |\nabla u|^2)^{p/2} - \varepsilon^p}{p} \, dx - \langle f, u \rangle, \quad \varepsilon > 0. \quad (19)$$

On $W^{1,p}(\Omega)$, define the reaction energy

$$\mathcal{E}_{p,R}(u) = \int_{\Omega} \left(\frac{1}{p} |\nabla u|^p + \frac{1}{p} |u|^p \right) \, dx - \langle f, u \rangle. \quad (20)$$

For a differentiable convex functional F , its Bregman divergence is

$$D_F(u, v) = F(u) - F(v) - \langle F'(v), u - v \rangle \geq 0. \quad (21)$$

It measures the part of the energy increment not accounted for by the tangent functional at v . In general $D_F(u, v) \neq D_F(v, u)$, so it is not a metric. The same notation is used pointwise for a convex density: $D_\phi(a, b) = \phi(a) - \phi(b) - \nabla\phi(b) \cdot (a - b)$.

For scalar arguments set $V(s) = |s|^{(p-2)/2} s$, and for vector arguments set

$$V(\xi) = |\xi|^{(p-2)/2} \xi, \quad V_\varepsilon(\xi) = (\varepsilon^2 + |\xi|^2)^{(p-2)/4} \xi.$$

The corresponding natural distances are

$$\begin{aligned} d_V(u, v)^2 &= \|V(\nabla u) - V(\nabla v)\|_{L^2}^2, \\ d_{V,\varepsilon}(u, v)^2 &= \|V_\varepsilon(\nabla u) - V_\varepsilon(\nabla v)\|_{L^2}^2, \\ d_{V,R}(u, v)^2 &= d_V(u, v)^2 + \|V(u) - V(v)\|_{L^2}^2. \end{aligned}$$

We use $\mathcal{E}_\bullet$ and $d_{V,\bullet}$ only as paired abbreviations for one of $(\mathcal{E}_p, d_V)$, $(\mathcal{E}_{p,\varepsilon}, d_{V,\varepsilon})$, or $(\mathcal{E}_{p,R}, d_{V,R})$. The scalar map applies to the reaction term and the vector map to gradients. The map V turns the weighted quadratic structure of a p -operator into an L^2 -type distance (Barrett and Liu, 1993; Diening and Kreuzer, 2008; Lee and Park, 2025).

Throughout this section, $A \lesssim_p B$ means $A \leq C(p)B$, and $A \simeq_p B$ means that both inequalities hold with positive constants depending only on p ; a further subscript lists any additional permitted dependence.

Lemma 4.5 (Pointwise p -structure). *For $p \geq 2$, constants depending only on p satisfy, for every $a, b \in \mathbb{R}^d$,*

$$\frac{|a|^p}{p} - \frac{|b|^p}{p} - |b|^{p-2} b \cdot (a - b) \simeq_p (|a| + |b|)^{p-2} |a - b|^2 \simeq_p |V(a) - V(b)|^2, \quad (22)$$

$$|a - b|^p \lesssim_p |V(a) - V(b)|^2. \quad (23)$$

The analogous statements hold for the regularized density and V_ε , with $\varepsilon > 0$ fixed.

Proof. For $\phi(\xi) = |\xi|^p/p$, write $D_\phi(a, b)$ as

$$\int_0^1 (1-t)(a-b)^T \nabla^2 \phi(b + t(a-b))(a-b) dt.$$

The radial and tangential eigenvalues of $\nabla^2 \phi(\xi)$ are comparable to $|\xi|^{p-2}$. Segment integration gives the first weighted quadratic equivalence, and the standard monotonicity estimate for $|\xi|^{p-2} \xi$ gives the V -map equivalence. Since $(|a| + |b|)^{p-2} \geq |a - b|^{p-2}/2^{p-2}$, Equation (23) follows. Replacing $|\xi|$ by $(\varepsilon^2 + |\xi|^2)^{1/2}$ gives the regularized case. These estimates are also the basis of the natural-error analysis in Barrett and Liu (1993); Diening and Kreuzer (2008). $\square$

Theorem 4.6 (Energy–natural-distance equivalence). *For each of the three energy–space pairs*

$$(\mathcal{E}_p, X_{p,\circ}), \quad (\mathcal{E}_{p,\varepsilon}, X_{p,\circ}), \quad (\mathcal{E}_{p,R}, W^{1,p}(\Omega)),$$

let u^ be the minimizer. Then, for every admissible u , respectively,*

$$\mathcal{E}_p(u) - \mathcal{E}_p(u^*) \simeq_p d_V(u, u^*)^2, \quad (24)$$

$$\mathcal{E}_{p,\varepsilon}(u) - \mathcal{E}_{p,\varepsilon}(u^*) \simeq_{p,\varepsilon} d_{V,\varepsilon}(u, u^*)^2, \quad (25)$$

$$\mathcal{E}_{p,R}(u) - \mathcal{E}_{p,R}(u^*) \simeq_p d_{V,R}(u, u^*)^2. \quad (26)$$

In the corresponding spaces,

$$\|u - u^*\|_{W^{1,p}}^p \lesssim_{p,\Omega,\varepsilon} d_{V,\bullet}(u, u^*)^2, \quad (27)$$

where the constants in the pure and reaction cases are independent of ε .

Proof. The first-order condition for the minimizer on its feasible space is $\langle \mathcal{E}'_\bullet(u^*), u - u^* \rangle = 0$. Hence the energy gap is exactly the Bregman divergence of the nonlinear density at u^* ; the linear load cancels. Integrating Lemma 4.5 over gradients gives Equation (24); its regularized counterpart gives Equation (25). For the reaction energy, apply the same scalar inequality to u in addition to the gradient inequality, which proves Equation (26). Finally, Equation (23) controls the L^p gradient difference. The zero-mean Poincaré inequality controls the function difference for the pure and regularized models, while the scalar reaction term directly controls it in Equation (20). This proves Equation (27). $\square$

Corollary 4.7 (Metric exponents). *Fix any one of the three energy–space–distance triples in Theorem 4.6. If a sequence G_m in that feasible space satisfies*

$$\mathcal{E}_\bullet(G_m) - \mathcal{E}_\bullet(u^*) = O(m^{-\alpha}) \quad (28)$$

for some $\alpha > 0$, then

$$d_{V,\bullet}(G_m, u^*) = O(m^{-\alpha/2}), \quad \|G_m - u^*\|_{W^{1,p}} = O(m^{-\alpha/p}). \quad (29)$$

Proof. The lower bounds in Equations (24) to (26) give $d_{V,\bullet}^2 \lesssim m^{-\alpha}$, and taking a square root proves the first estimate. Combining the same bound with Equation (27) and taking a p -th root proves the second. Because $\varepsilon > 0$ is fixed in the regularized model and all iterates lie in the same feasible space, the implicit constants do not depend on m . $\square$

Remark 4.8 (Scope). The corollary is restricted to the three standard p -Laplacian-type energies just defined. Linear, cubic, and hyperbolic-sine energies have their own quadratic or local-Hessian geometry and are not instances of this statement merely because they are convex. Conversely, an unspecified energy with p -growth need not satisfy the Bregman– V equivalence in Lemma 4.5.

5 Conditional convergence and finite-trajectory projection bounds

We distinguish descent for the implemented nonlinear method from approximation in a frozen Hilbert metric. The first involves continuous-dictionary coverage, score accuracy, and coefficient correction. The second concerns the nested spaces generated by the algorithm and does not require the nonlinear method to maximize a frozen score. The two are connected only conditionally, and for the quartic energies the connection can also be expressed by an exact identity.

Throughout, m counts accepted independent directions. A rejected attempt or a direction already in the active span does not increment this index. The starting index is m_0 and $j = m - m_0$. We use $\widehat{G}_m$ for a computed state, $G_m^\sharp$ for an exact active-space energy minimizer, and z_m for a frozen orthogonal projection. These states need not coincide. All derivatives below act on the specified feasible linear space, so that $\mathcal{E}'(u^*) = 0$ there. Affine constraints are reduced to this setting by a fixed feasible lift.

5.1 Finite-pool descent and the residual-comparison condition

Let X be a real Banach space, $\mathcal{E} : X \rightarrow \mathbb{R}$ a differentiable convex functional on the states and segments used below, and u^* a minimizer. The symmetric dictionary $\mathcal{D}_X$ consists of X -unit vectors. Set

$$\widehat{\delta}_m = \mathcal{E}(\widehat{G}_m) - \mathcal{E}(u^*), \quad S_m^X = \sup_{g \in \mathcal{D}_X} \left| \langle \mathcal{E}'(\widehat{G}_m), g \rangle \right|.$$

The exact finite-pool score is $s_m(g) = |\langle \mathcal{E}'(\widehat{G}_m), g \rangle|$. The available computed scores satisfy

$$|\widetilde{s}_m(g) - s_m(g)| \leq \xi_m \quad (g \in \mathcal{P}_m), \quad \widetilde{s}_m(g_{m+1}) \geq t_m^{\text{pool}} \max_{g \in \mathcal{P}_m} \widetilde{s}_m(g), \quad (30)$$

where $\mathcal{P}_m \subset \mathcal{D}_X$ is finite and $0 < t_m^{\text{pool}} \leq 1$. When $S_m^X > 0$, define its continuous coverage by

$$\tau_m = \frac{\max_{g \in \mathcal{P}_m} s_m(g)}{S_m^X} \in [0, 1]. \quad (31)$$

Set $\tau_m = 1$ when $S_m^X = 0$. The selection weakness t_m^{pool} and the coverage τ_m are different quantities.

Assume that a common $L > 0$ satisfies

$$\mathcal{E}(\widehat{G}_m + \lambda g) \leq \mathcal{E}(\widehat{G}_m) + \lambda \langle \mathcal{E}'(\widehat{G}_m), g \rangle + \frac{L}{2} \lambda^2, \quad |\lambda| \leq 1, \quad g \in \mathcal{D}_X. \quad (32)$$

The expanded space contains $\widehat{G}_m + \lambda g_{m+1}$, and the correction satisfies

$$\mathcal{E}(\widehat{G}_{m+1}) \leq \inf_{v \in \widehat{\Phi}_{m+1}} \mathcal{E}(v) + \varepsilon_m^{\text{opt}}, \quad \varepsilon_m^{\text{opt}} \geq 0. \quad (33)$$

This assumption concerns the true energy. A numerical objective needs its own comparison with that energy.

Proposition 5.1 (Descent with a resolved score). *Let $a_m \geq 0$ be any valid lower bound for $s_m(g_{m+1})$. Under (32)–(33),*

$$\widehat{\delta}_{m+1} \leq \widehat{\delta}_m - H_L(a_m) + \varepsilon_m^{\text{opt}}. \quad (34)$$

Here H_L is the directional-decrease function defined in Section 3. In particular, (30) gives the valid lower bound

$$a_m = \left[t_m^{\text{pool}} \tau_m S_m^X - (1 + t_m^{\text{pool}}) \xi_m \right]_+. \quad (35)$$

Proof. The triangle inequality gives $s_m(g_{m+1}) \geq t_m^{\text{pool}} \max_{\mathcal{P}_m} s_m - (1 + t_m^{\text{pool}}) \xi_m$. Test the correction with a signed dictionary step in the true descent direction of length $\min\{1, a_m/L\}$. Both signs lie in the expanded linear space. Minimizing the quadratic upper bound on that segment gives $H_L(a_m)$ and proves (34). $\square$

The radial stationarity error enters when an atomic source is used to bound S_m^X . Write $\mathcal{K}_1(\mathcal{D})$ for the closed absolutely convex hull of a bounded dictionary. If $u^* \in B_{\text{src}} \mathcal{K}_1(\mathcal{D}_X)$ with $B_{\text{src}} > 0$ and $|\langle \mathcal{E}'(\widehat{G}_m), \widehat{G}_m \rangle| \leq \eta_m^{\text{rad}}$, convexity implies

$$\widehat{\delta}_m \leq \eta_m^{\text{rad}} + B_{\text{src}} S_m^X. \quad (36)$$

Consequently one may instead use

$$a_m = \left[t_m^{\text{pool}} \tau_m \frac{\widehat{\delta}_m - \eta_m^{\text{rad}}}{B_{\text{src}}} - (1 + t_m^{\text{pool}}) \xi_m \right]_+.$$

This is a source-dependent estimate, and it is not an additional error term to be added after the score estimate (35). In the error framework of (Dereventsov and Temlyakov, 2022, Section 3, Theorem 3.2 and Lemma 6.3), energy correction and radial biorthogonality are separate errors. Their power-smoothness rate assumes uniform positive continuous weakness and prescribed decay of those errors. Here they correspond to $\varepsilon_m^{\text{opt}}$ and η_m^{rad} , with the latter at the preceding state in (36). Pool coverage and score quadrature determine a further effective weakness through (35). Accuracy of the computed correction alone does not ensure continuous weakness.

Lemma 5.2 (A nonautonomous scalar bound). *Suppose $x_m \geq 0$ and $x_{m+1} \leq x_m - b_m x_m^q$ for $q > 1$ and $b_m \geq 0$. For $x_0 > 0$,*

$$x_n \leq \left[x_0^{1-q} + (q-1) \sum_{m=0}^{n-1} b_m \right]^{-1/(q-1)}. \quad (37)$$

If $x_0 = 0$, the sequence vanishes. In particular, $b_m \geq b(m+1)^{-s}$, $b > 0$, $0 \leq s < 1$, implies $x_n = O((n+1)^{-(1-s)/(q-1)})$.

Proof. As long as $x_{m+1} > 0$, convexity gives $x_{m+1}^{1-q} - x_m^{1-q} \geq (q-1)x_m^{-q}(x_m - x_{m+1}) \geq (q-1)b_m$. Sum. Once a term vanishes, nonnegativity and the recurrence force all later terms to vanish. The power estimate follows by comparison of the sum with an integral. $\square$

Theorem 5.3 (Conditional residual-comparison rate). *Let $p > 2$. Suppose, at the computed states,*

$$\widehat{\delta}_m \geq c_U \left\| \widehat{G}_m - u^* \right\|_X^p, \quad c_U > 0, \quad (38)$$

$$S_m^X \geq \kappa_0(m+1)^{-\chi} \left\| \mathcal{E}'(\widehat{G}_m) \right\|_{X^*}, \quad \kappa_0 > 0, \quad 0 \leq \chi < \frac{1}{2}. \quad (39)$$

Assume (30)–(33), and, uniformly in m ,

$$t_m^{\text{pool}} \tau_m \geq \underline{t} > 0, \quad (1 + t_m^{\text{pool}}) \xi_m \leq \vartheta t_m^{\text{pool}} \tau_m S_m^X, \quad 0 \leq \vartheta < 1.$$

With a_m as in (35), assume also $\varepsilon_m^{\text{opt}} \leq \zeta H_L(a_m)$, $0 \leq \zeta < 1$. Then

$$\widehat{\delta}_m = O((m+1)^{-p(1-2\chi)/(p-2)}). \quad (40)$$

The constant depends on $p, c_U, \kappa_0, \underline{t}, L, \vartheta, \zeta$ and $\widehat{\delta}_0$. For the exact continuous method take $\tau_m = 1$, $\xi_m = \varepsilon_m^{\text{opt}} = 0$.

Proof. Convexity and (38) give

$$\widehat{\delta}_m \leq \left\| \mathcal{E}'(\widehat{G}_m) \right\|_{X^*} (\widehat{\delta}_m / c_U)^{1/p}.$$

For positive error this yields gradient domination. Thus $a_m \geq b_m^* := K(m+1)^{-\chi} \widehat{\delta}_m^{(p-1)/p}$, where $K = (1 - \vartheta) \underline{t} \kappa_0 c_U^{1/p}$. Descent and absorption imply monotonicity, so $b_m^* \leq A := K \widehat{\delta}_0^{(p-1)/p}$. For $0 \leq b \leq A$, $H_L(b) \geq b^2 / (2 \max\{L, A\})$. Since H_L is increasing, (34) implies

$$\widehat{\delta}_{m+1} \leq \widehat{\delta}_m - \frac{(1 - \zeta) K^2}{2 \max\{L, A\}} (m+1)^{-2\chi} \widehat{\delta}_m^{2(p-1)/p}.$$

Apply Lemma 5.2. The case $\widehat{\delta}_0 = 0$ is immediate from descent and absorption. $\square$

This theorem does not assert that density of $\text{span } \mathcal{D}_X$ implies (39). For relatively compact dictionaries in an infinite-dimensional space, a uniform norm comparison on the entire dual space is generally impossible. The comparison is therefore a hypothesis on the specified residual trajectory. The resulting estimate improves on the atomic energy baseline m^{-1} exactly when $\chi < 1/p$.

5.2 Energy and the natural distance

For the standard pure, regularized, and reaction p -energies, the energy–distance equivalence and the $W^{1,p}$ conversion are established in Theorem 4.6 and Corollary 4.7. Therefore, if the conditional energy estimate in Theorem 5.3 is $\widehat{\delta}_m = O(m^{-\alpha})$, then

$$d_{V,\bullet}(\widehat{G}_m, u^*) = O(m^{-\alpha/2}), \quad \|\widehat{G}_m - u^*\|_{W^{1,p}} = O(m^{-\alpha/p}),$$

under the corresponding zero-mean or reaction assumptions. An upper energy bound gives an upper metric bound; it is not an equality with a fitted slope.

Thus (40) has natural-distance exponent $p(1-2\chi)/(2(p-2))$ and Sobolev exponent $(1-2\chi)/(p-2)$ for the pure, regularized, and reaction energies. The scalar comparison in Theorem 5.3 does not automatically extend to other energy densities.

5.3 Local Hessian transfer and the same-dictionary requirement

Previous theoretical work provides substantial support for the transfer argument. Xu–Xu (Xu and Xu, 2025, Theorem 3.1, equations (17)–(19)) analyze exact active-space minimization of a convex, L -smooth functional in a Sobolev Hilbert space. In their notation, the continuous selection strengths γ_i , source radius B , and innovations $\bar{g}_i = (I - P_{i-1})g_i$ yield

$$\delta_i \leq \delta_{i-1} - \frac{\gamma_i^2 \delta_{i-1}^2}{2LB^2 \|\bar{g}_i\|^2}.$$

Their ReLU dictionary estimate then gives an energy rate $n^{-1-(2(k-s)+1)/d}$ for constant positive weakness in H^s , with $k \geq s$; strong convexity gives the corresponding norm bound. Here s is the Sobolev order, not the iteration index. That exponent belongs to their normalized dictionary and hypotheses; it is not imported into the present finite, discretely normalized pool without a separate compatible entropy proof. The argument below states explicit sufficient conditions for transferring nonlinear selection to a frozen metric. It does not establish a new general nonlinear greedy rate. Global smoothness, exact correction, and the dictionary assumptions in the direct theorem cannot be inferred from a finite-pool computation or from local Hessian positivity.

Relative to Dereventsov–Temlyakov, the finite-pool proposition below separates coverage, score-quadrature, radial-stationarity, and coefficient-correction errors. Relative to Xu–Xu, the transfer theorem retains their exact active-space and entropy mechanism and adds only a conditional local Hessian transfer. Relative to Li–Siegel, the entropy lemma specializes the volume argument to weak selection and unrenormalized projected atoms. These comparisons identify the decomposition of the selection, quadrature, stationarity, and correction terms used in the present transfer argument; they do not by themselves establish a new nonlinear rate.

Let H be a real Hilbert space on which $\mathcal{E}$ is twice Fréchet differentiable in an open convex neighborhood of u^* . Assume $A_* = \mathcal{E}''(u^*)$ is bounded, symmetric, and coercive on H . Write $a_*(v, w) = \langle A_* v, w \rangle$ and $\|v\|_*^2 = a_*(v, v)$. Suppose the closed ball of radius R_* around u^* lies in that neighborhood and

$$|[\mathcal{E}''(u) - A_*]v, w| \leq L_* \|u - u^*\|_* \|v\|_* \|w\|_*, \quad \|u - u^*\|_* \leq R_*, \quad L_* R_* \leq \frac{1}{2}. \quad (41)$$

For finite-dimensional $\Phi \subset H$ let z_Φ be the frozen orthogonal projection of u^* and $G_\Phi^\sharp$ an exact minimizer of $\mathcal{E}$ over Φ . Assume both points lie in this ball.

Lemma 5.4 (Taylor and projection estimates). *Under (41), for u in the ball,*

$$\left| \mathcal{E}(u) - \mathcal{E}(u^*) - \frac{1}{2} \|u - u^*\|_*^2 \right| \leq \frac{L_*}{6} \|u - u^*\|_*^3, \quad (42)$$

$$\|\mathcal{E}'(u) - A_*(u - u^*)\|_{*'} \leq \frac{L_*}{2} \|u - u^*\|_*^2, \quad (43)$$

$$\left\| G_\Phi^\sharp - z_\Phi \right\|_* \leq L_* \|z_\Phi - u^*\|_*^2. \quad (44)$$

Here the dual norm in (43) is taken over H .

Proof. One and two integrations of (41) give the first two estimates. Put $d = G_\Phi^\sharp - z_\Phi \in \Phi$. Use $\langle \mathcal{E}'(G_\Phi^\sharp), d \rangle = 0$ and $a_*(z_\Phi - u^*, d) = 0$. Subtracting A_*d along the segment from z_Φ to $G_\Phi^\sharp$ bounds $\|d\|_*^2$ by $L_* R_* \|d\|_*^2 + (L_*/2) \|z_\Phi - u^*\|_*^2 \|d\|_*$. Absorb the first term. All segments remain in the ball by convexity. $\square$

Assume $\mathcal{D}_X \subset H$ and the algorithmic dictionary satisfies the uniform norm comparison

$$0 < c_* \leq \|g\|_* \leq C_* < \infty \quad (g \in \mathcal{D}_X), \quad \mathcal{D}_* = \{g / \|g\|_* : g \in \mathcal{D}_X\}. \quad (45)$$

Let Φ_m be nested finite-dimensional spaces with $\Phi_{m+1} = \Phi_m + \text{span}\{g_{m+1}\}$, where each selected direction is independent of Φ_m . Denote their frozen orthogonal projections by P_m , and set $z_m = P_m u^*$, $r_m = \|u^* - z_m\|_*$, and $G_m^\sharp = G_{\Phi_m}^\sharp$. At an exact active-space state set

$$S_m^* = \sup_{d \in \mathcal{D}_*} |a_*(z_m - u^*, d)|, \quad \eta_m^H = \sup_{g \in \mathcal{D}_X} |\langle \mathcal{E}'(G_m^\sharp), g \rangle - a_*(z_m - u^*, g)|.$$

The preceding lemma yields $\eta_m^H \leq C_T \|z_m - u^*\|_*^2$, with a constant uniform on the specified ball and dictionary. This absolute bound alone need not be small compared with S_m^* .

Proposition 5.5 (Selection under a relative score condition). *Suppose nonlinear selection over $\mathcal{D}_X$ has strength $t \in (0, 1]$. For $0 < t_0 < tc_*/C_*$, the condition*

$$(1 + t)\eta_m^H \leq (tc_* - t_0 C_*) S_m^* \quad (46)$$

implies $|a_(z_m - u^*, g_{m+1} / \|g_{m+1}\|_*)| \geq t_0 S_m^*$.*

Proof. The selected unnormalized frozen score is at least $t \sup_{g \in \mathcal{D}_X} |a_*(z_m - u^*, g)| - (1 + t)\eta_m^H$. The supremum is at least $c_* S_m^*$, and division by the selected norm costs at most C_* . This proves the assertion. $\square$

Fix m_0 , $Q_0 = I - P_{m_0}$ and

$$H_0 = Q_0 H, \quad f_0 = Q_0 u^*, \quad \mathcal{D}_{\text{rst}} = Q_0 \mathcal{D}_* \setminus \{0\}.$$

The projected atoms have norm at most one and are not renormalized. The entropy numbers use the 2^{n-1} -ball convention fixed in the variational setting; consequently a cover by 2^j balls in the volume argument below is represented by ε_{j+1} , not by a second definition of the same symbol.

Lemma 5.6 (Entropy bound for bounded-atom weak OGA). *Let $\mathcal{D}_0$ be symmetric with atom norms at most one in a Hilbert space, $K = \mathcal{K}_1(\mathcal{D}_0)$ compact, and $f_0 \in B_* K$ with $B_* \geq 0$. Let weak OGA of strength $t_0 > 0$ start from the zero space. After $j \geq 1$ independent extensions its error satisfies*

$$\|f_0 - P_j f_0\| \leq \frac{B_* (j! V_j)^{1/j}}{t_0 \sqrt{j}} \varepsilon_{j+1}(K), \quad V_j = \frac{\pi^{j/2}}{\Gamma(j/2 + 1)}. \quad (47)$$

Proof. If $B_* = 0$, then $f_0 = 0$ and the assertion is immediate. Suppose $B_* > 0$. Write $y_i = \|f_0 - P_i f_0\|^2$ and $l_i = \|(I - P_{i-1})d_i\| > 0$. Orthogonality and the source condition give the best score at least y_{i-1}/B_* . Hence $y_i \leq y_{i-1} - t_0^2 y_{i-1}^2 / (B_*^2 l_i^2)$. If the error reaches zero the result is immediate. Otherwise taking reciprocals and using the arithmetic–geometric mean inequality yields

$$y_j \leq \frac{B_*^2}{t_0^2 \sum_{i=1}^j l_i^{-2}} \leq \frac{B_*^2}{t_0^2 j} \left(\prod_{i=1}^j l_i \right)^{2/j}.$$

The cross-polytope generated by $\{\pm d_1, \dots, \pm d_j\}$ has j -dimensional volume $(2^j/j!) \prod_i l_i$ and lies in the orthogonal projection of K onto their span. A cover of K by 2^j balls of radius $\varepsilon_{j+1}(K) + h$, $h > 0$, projects to a cover of that cross-polytope. Volume comparison gives $\prod_i l_i \leq j! V_j(\varepsilon_{j+1}(K) + h)^j$. Let $h \downarrow 0$ and substitute. No projected atom was rescaled. $\square$

The argument is the volume/entropy argument of Li–Siegel, specialized here to weak selection and to atoms bounded by one. It corresponds to Theorem 4.2 and equation (4.4) of Li and Siegel (2024), with the projection recurrence in equations (4.5)–(4.9); the numbering is that of the cited version of the reference. The specialization records the innovation weights of the restarted spaces; it is not an independent entropy theorem. The factorial–volume factor divided by $\sqrt{j}$ is bounded uniformly in j by Stirling’s formula.

Theorem 5.7 (Conditional Hessian transfer after restart). *Assume (41) and (45). For every state in $m_0 \leq m \leq m_0 + J$, suppose that the exact nonlinear state and frozen projection lie in the specified ball. For each selection step $m_0 \leq m < m_0 + J$, assume nonlinear selection has strength at least t , and (46) holds with the same $t_0 > 0$. On the unrenormalized restarted dictionary assume*

$$f_0 \in B_* \mathcal{K}_1(\mathcal{D}_{\text{rst}}), \quad \varepsilon_j(\mathcal{K}_1(\mathcal{D}_{\text{rst}}); H_0) \leq C_{\text{ent}} j^{-\beta}, \quad (48)$$

for $1 \leq j \leq J + 1$, with compact closed absolutely convex hull, $B_*, C_{\text{ent}} > 0$, and $\beta > 0$. Then

$$\|z_{m_0+j} - u^*\|_* \leq C B_* C_{\text{ent}} t_0^{-1} j^{-\beta}, \quad \mathcal{E}(G_{m_0+j}^\#) - \mathcal{E}(u^*) \leq C_{\text{nl}} j^{-2\beta}.$$

Here C is an absolute constant and C_{nl} additionally depends on $L_*, R_*, B_*, C_{\text{ent}}, t_0$. The $j + 1$ index in Lemma 5.6 is the harmless consequence of the global 2^{n-1} -ball convention; the entropy assumption is stated through $J + 1$ for this reason. For a finite J the conclusion is restricted to that finite range.

Proof. Proposition 5.5 gives weak frozen selection. The residual is orthogonal to the starting space, so its scores against d and $Q_0 d$ agree. The spaces after the starting index therefore form weak OGA on H_0 with dictionary $\mathcal{D}_{\text{rst}}$. Apply Lemma 5.6. Lemma 5.4 gives $\|G_{m_0+j}^\# - u^*\|_* \leq r_{m_0+j} + L_* r_{m_0+j}^2 \leq (1 + L_* R_*) r_{m_0+j}$, and (42) gives the energy bound. $\square$

Constants uniform in a terminal width are required for an asymptotic conclusion. No exponent depending on a ReLU degree is substituted without a proof for this dictionary.

This is a conditional transfer statement. The family below verifies coercivity, source membership, and compactness, but it does not verify (46) or a quantitative entropy exponent for $\lambda > 0$. Hence the theorem does not yield an unconditional nonlinear asymptotic rate for that family.

A family satisfying the structural assumptions. Take $H = H^1(0, 1)$ and

$$\mathcal{E}_\lambda(u) = \int_0^1 \left(\frac{1}{2} |u'|^2 + \frac{1}{2} u^2 + \frac{\lambda}{4} u^4 \right) dx - \ell(u), \quad \lambda \geq 0.$$

For a fixed integer $k \geq 2$ and $0 < a < b < 1$, put $g_t = (x - t)_+^k / \|(x - t)_+^k\|_{H^1}$, $t \in [a, b]$, and $\mathcal{D}_X = \{\pm g_t\}$. Choose any finite signed measure μ with total variation at most B_{src} , set $u^* = \int_a^b g_t d\mu(t)$, and manufacture $\ell(v) = \int (u^* v' + u^* v + \lambda (u^*)^3 v)$. The embedding $H^1(0, 1) \hookrightarrow L^\infty$ makes this load continuous and the Hessian locally Lipschitz on H^1 balls. The load is defined in variational form; it should not be relabelled as a homogeneous Neumann load unless the corresponding boundary-flux compatibility has also been checked. Its bilinear form is $\int (v' w' + (1 + 3\lambda (u^*)^2) v w)$, which is coercive in H^1 . The norm comparison (45) follows, the dictionary is compact, and $u^* \in C_* B_{\text{src}} \mathcal{K}_1(\mathcal{D}_*)$. Its closed absolutely convex hull is compact: finite nets for the dictionary approximate that hull by bounded finite-dimensional convex hulls. Projection by Q_0 preserves the source and cannot increase entropy. Thus these structural assumptions are simultaneously meaningful for an infinite family of targets, not only for one atom or one fixed finite space. For $\lambda > 0$, the relative-score condition (46) and a quantitative entropy exponent remain additional hypotheses. For $\lambda = 0$, $\eta_m^H = 0$ and the setting reduces to ordinary Hilbert OGA. This example does not, by itself, establish a new nonlinear fast rate.

The missing relative-score condition can nevertheless be reduced to a quantitative small-nonlinearity check. Let $C_{\text{emb}} = \|H^1(0, 1) \hookrightarrow L^\infty(0, 1)\|$ (for the standard squared H^1 norm one may use the explicit bound $C_{\text{emb}} \leq \sqrt{2}$) and $M_* = \|u^*\|_{L^\infty}$, and use the H^1 -normalized atoms above. Since a_* dominates the standard H^1 inner product, the quartic Hessian satisfies, on the specified R_* -ball,

$$L_\lambda := 3\lambda C_{\text{emb}}(C_{\text{emb}} R_* + 2M_*), \quad L_\lambda R_* \leq \frac{1}{2}, \quad (49)$$

and the norm-comparison and source constants can be chosen as

$$c_* = 1, \quad C_\lambda = \sqrt{1 + 3\lambda M_*^2}, \quad B_\lambda = C_\lambda B_{\text{src}}. \quad (50)$$

Indeed, $\|g_t\|_* = \sqrt{1 + 3\lambda \|u^* g_t\|_{L^2}^2}$ lies between 1 and C_λ , and the change of variables from g_t to $g_t/\|g_t\|_*$ multiplies the atomic budget by at most C_λ . Put

$$C_{T,\lambda} = L_\lambda \left[1 + \frac{1}{2}(1 + L_\lambda R_*)^2 \right]. \quad (51)$$

The projection estimates then give $\eta_m^H \leq C_\lambda C_{T,\lambda} r_m^2$. Consequently the following explicit bound is sufficient for the relative-score condition:

$$\Delta_\lambda := (tc_* - t_0 C_\lambda) - (1 + t) C_\lambda^2 C_{T,\lambda} B_{\text{src}} > 0. \quad (52)$$

This is a check on specified constants, not a replacement for the invariant ball or entropy arguments. For fixed R_* , M_* , B_{src} , t , and t_0 , the left-hand nonlinear term tends to zero as $\lambda \downarrow 0$; continuity therefore gives a positive, specified λ_0 whenever $t_0 < tc_*$, provided the constants are held fixed. The admissible interval $0 < \lambda \leq \lambda_0$ must satisfy both (49) and (52) throughout the interval, with the values of R_* , M_* , B_{src} , t , and t_0 specified.

Sufficient conditions for the concrete semilinear family. The following statement isolates the sufficient conditions needed for a nonlinear claim. Compactness of the parameter interval alone does not supply any of their quantitative parts. Fix m_0 , $Q_0 = I - P_{m_0}$, and the unrenormalized restarted dictionary $\mathcal{D}_{\text{rst}} = Q_0 \mathcal{D}_* \setminus \{0\}$, where $\mathcal{D}_* = g_t/\|g_t\|_* : t \in [a, b]$ with both signs included. The sufficient conditions on a finite range $0 \leq j \leq J$ are:

- (A1) an initial radius and a specified R_* for which every exact active minimizer, frozen projection, and corrected state in the stated finite range stays in $B_*(u^*, R_*)$; for a specified $r_{\text{ent}} > 0$ this includes $r_{m_0} \leq r_{\text{ent}}$ and $L_\lambda R_* \leq \frac{1}{2}$;
- (A2) a uniform selection strength t and $t_0 < tc_*/C_\lambda$ for which the displayed bound Δ_λ is positive at every stated step (or an explicitly stated finite-prefix exception);
- (A3) a quantitative entropy estimate for this same projected, unrenormalized object, with the ball-counting convention fixed above,

$$\varepsilon_n(\mathcal{K}_1(\mathcal{D}_{\text{rst}}); H_0) \leq C_{\text{ent}} n^{-\beta_A}, \quad 1 \leq n \leq J+1, \quad (53)$$

and $Q_0 u^* \in B_\lambda \mathcal{K}_1(\mathcal{D}_{\text{rst}})$; and

- (A4) correction, quadrature, and radial errors absorbed in the finite-pool score bound, with tolerances prescribed from the available finite-pool quantities.

For $k \geq 2$, one possible target is $\beta_A = 1$, which would follow from a direct parameter-metric estimate for $t \mapsto g_t$ together with a covering argument for the absolutely convex hull. Until such an estimate is proved for the same projected dictionary, only an exponent established by that argument may be used. If the estimate is available only for a fixed J , the resulting conclusion is restricted to that finite range.

Corollary 5.8 (Concrete-family conditional rate). *Assume the sufficient conditions above hold for some $\lambda \in (0, \lambda_0]$ and $0 < \beta_A$, and use exact active-space minimization in the statement of the rate. Then the restarted CGA trajectory satisfies, for $1 \leq j \leq J$,*

$$\|z_{m_0+j} - u^*\|_* \leq C B_\lambda C_{\text{ent}} t_0^{-1} j^{-\beta_A}, \quad \mathcal{E}_\lambda(G_{m_0+j}^\#) - \mathcal{E}_\lambda(u^*) \leq C_{\text{nl}} j^{-2\beta_A}. \quad (54)$$

The constants are those of Theorem 5.7; if the conditions are established only through J , (54) is not an asymptotic assertion. An unconditional rate requires the relative-score, invariant-neighbourhood, and entropy conditions to hold uniformly along the trajectory, with constants independent of J .

Remark 5.9 (Scope of the conditional family result). The family construction and the explicit Hessian and norm-comparison formulas reduce the abstract hypotheses to quantities that can in principle be estimated. They do not by themselves establish a positive relative-score bound, the entropy estimate in (53), or invariance of a computed trajectory. These remain assumptions to be verified before the conditional theorem can be stated as an unconditional rate result.

Scope of the function-space assumptions. Full normalization of boundary-concentrating truncated powers need not give a compact dictionary. For example, as $t \uparrow 1$, the function

$$b_t(x) = \frac{(x-t)_+^k}{\|(x-t)_+^k\|_{H^1(0,1)}}$$

has H^1 norm one but L^2 norm tending to zero. A strongly convergent H^1 subsequence would therefore have limit zero, a contradiction. For the two-dimensional sinh model, an H^1 ball does not give uniform L^∞ bounds; a Hessian estimate restricted to bounded states is not a proof of (41) on an H^1 open ball. Without a separate functional-analytic argument we use that model only in the finite-dimensional trajectory setting. Degenerate pure p -energies likewise need not satisfy full-space Hessian coercivity.

5.4 Finite-reference scores and restricted stationarity

On a specified finite-dimensional space Y_M with a positive frozen inner product, assume $u^* \in Y_M$ and $\widehat{\Phi}_m \subset Y_M$. In this setting let z_m be the frozen projection of u^* onto $\widehat{\Phi}_m$. Let the fixed reference set $\mathcal{A}_M \subset Y_M$ contain all pools on the stated finite range. Assume $0 < c_M \leq \kappa_g := \|g\|_* \leq C_M < \infty$. Set

$$h_m(g) = |a_*(u^* - z_m, g)|, \quad S_m^{\text{val}} = \max_{g \in \mathcal{A}_M} h_m(g)/\kappa_g, \quad \tau_m^{\text{val}} = \frac{\max_{g \in \mathcal{P}_m} h_m(g)/\kappa_g}{S_m^{\text{val}}}$$

when $S_m^{\text{val}} > 0$. If the computed scores obey $|\tilde{s}_m(g) - h_m(g)| \leq \Delta_m$ on $\mathcal{P}_m$, the proof of Proposition 5.5 gives

$$\frac{h_m(g_{m+1})}{\kappa_{g_{m+1}}} \geq \frac{t_m^{\text{pool}} c_M \tau_m^{\text{val}} S_m^{\text{val}} - (1 + t_m^{\text{pool}}) \Delta_m}{C_M}. \quad (55)$$

Thus the following nonnegative lower bound $(t_m^{\text{pool}} c_M \tau_m^{\text{val}} - t_0 C_M) S_m^{\text{val}} - (1 + t_m^{\text{pool}}) \Delta_m$ suffices for weakness t_0 relative to this finite reference set. A negative value of this lower bound yields no conclusion about the actual selected score. A continuous conclusion additionally requires $S_m^{\text{val}} \geq (1 - \zeta_M) S_m^*$ with $0 \leq \zeta_M < 1$. Comparing two quadrature rules supplies a comparison of quadrature sensitivity, not an established bound for Δ_m .

The active-space KKT residual in a reference norm $\|\cdot\|_{0,M}$ is

$$k_m = \sup_{\substack{v \in \widehat{\Phi}_m \\ \|v\|_{0,M}=1}} |\langle \mathcal{E}'(\widehat{G}_m), v \rangle|. \quad (56)$$

The supremum is over the active space, even when its norm is inherited from Y_M ; set $k_m = 0$ if $\widehat{\Phi}_m = \{0\}$. If $\mathcal{E}$ is $\mu_{\text{act}} > 0$ strongly convex on the active space and its exact minimizer exists, then

$$\|\widehat{G}_m - G_m^\# \|_{0,M} \leq k_m / \mu_{\text{act}}, \quad \mathcal{E}(\widehat{G}_m) - \mathcal{E}(G_m^\#) \leq k_m^2 / (2\mu_{\text{act}}).$$

The first follows by strong monotonicity paired with the state difference; the second follows by minimizing the strong-convexity quadratic. A bound C_{score} on $\|\mathcal{E}''(u)g|_{\widehat{\Phi}_m}\|_{\widehat{\Phi}_m^*}$ over the relevant segments and pool gives an additional score error at most $C_{\text{score}} k_m / \mu_{\text{act}}$. All norms and spaces must be specified before using such estimates. In particular,

$$\inf_{0 \neq v \in \widehat{\Phi}_m} \frac{a_*(v, v)}{\|v\|_{0,M}^2} \geq \inf_{0 \neq v \in Y_M} \frac{a_*(v, v)}{\|v\|_{0,M}^2}.$$

A positive active-space minimum does not establish a lower bound on Y_M or on an infinite-dimensional space.

5.5 An approximate-source estimate for the computed trajectory

Let Y be a finite-dimensional real Hilbert space containing u^* , all active spaces, and a fixed finite symmetric dictionary $\mathcal{D}_{\text{fin}}$ of frozen-unit vectors. For $m_0 \leq m < m_0 + J$ assume

$$\widehat{\Phi}_{m+1} = \widehat{\Phi}_m + \text{span}\{d_{m+1}\}, \quad d_{m+1} \in \mathcal{D}_{\text{fin}}, \quad d_{m+1} \notin \widehat{\Phi}_m. \quad (57)$$

The dictionary must contain every accepted direction in this finite range. Here P_m denotes the frozen orthogonal projection onto $\widehat{\Phi}_m$. Set

$$z_m = P_m u^*, \quad e_m = u^* - z_m, \quad r_m = \|e_m\|_*, \\ S_m^{\text{fin}} = \max_{d \in \mathcal{D}_{\text{fin}}} |a_*(e_m, d)|, \quad \ell_m = \|(I - P_m)d_{m+1}\|_* > 0.$$

Define $\theta_m = |a_*(e_m, d_{m+1})|/S_m^{\text{fin}}$ if $S_m^{\text{fin}} > 0$, and $\theta_m = 0$ otherwise. The vector $(I - P_m)d_{m+1}$ is the innovation of the candidate d_{m+1} , that is, its component orthogonal to $\hat{\Phi}_m$, and ℓ_m is the norm of that innovation. Let $Q_0 = I - P_{m_0}$ and assume explicit coefficients satisfy

$$v_B = \sum_{d \in \mathcal{D}_{\text{fin}}} a_d Q_0 d, \quad \sum_d |a_d| \leq B_{\text{entry}}, \quad \|Q_0 u^* - v_B\|_* \leq \sigma, \quad B_{\text{entry}} \geq 0. \quad (58)$$

The vectors $Q_0 d$ are not renormalized, and B_{entry} need not equal the global atomic radius B_{src} .

Theorem 5.10 (Weighted finite-trajectory source bound). *Under (57)–(58), take $0 \leq \underline{\theta}_m \leq \theta_m$ and set*

$$W_j = \sum_{m=m_0}^{m_0+j-1} \frac{\theta_m^2}{\ell_m^2}, \quad x_0 = (r_{m_0}^2 - \sigma^2)_+.$$

If $B_{\text{entry}} > 0$ and $x_0 > 0$, then for $0 \leq j \leq J$,

$$(r_{m_0+j}^2 - \sigma^2)_+ \leq \left(x_0^{-1} + \frac{W_j}{4B_{\text{entry}}^2} \right)^{-1}. \quad (59)$$

In particular, if $W_j > 0$, $r_{m_0+j}^2 \leq \sigma^2 + 4B_{\text{entry}}^2/W_j$. If $x_0 = 0$ or $B_{\text{entry}} = 0$, all later errors are at most σ . For $W_j = 0$ use projection monotonicity rather than division by zero.

Proof. For $w_m = (I - P_m)d_{m+1}$, Pythagoras gives

$$r_m^2 - r_{m+1}^2 = \frac{|a_*(e_m, d_{m+1})|^2}{\ell_m^2}. \quad (60)$$

The residual is orthogonal to both the starting and current spaces. Consequently

$$r_m^2 = a_*(e_m, Q_0 u^*) \leq B_{\text{entry}} S_m^{\text{fin}} + \sigma r_m, \quad S_m^{\text{fin}} \geq \frac{(r_m^2 - \sigma^2)_+}{2B_{\text{entry}}}$$

when $B_{\text{entry}} > 0$; the last step uses $2(r_m^2 - \sigma r_m) \geq r_m^2 - \sigma^2$. While both successive errors exceed σ , put $x_m = r_m^2 - \sigma^2 > 0$. Equation (60) implies $x_{m+1} \leq x_m - b_m x_m^2$ with $b_m = \underline{\theta}_m^2 / (4B_{\text{entry}}^2 \ell_m^2)$. Taking reciprocals proves (59) up to a crossing. After a crossing, monotonicity makes the positive part zero. If $S_m^{\text{fin}} = 0$ or $B_{\text{entry}} = 0$, the source pairing gives $r_m^2 \leq \sigma r_m$ directly. This also covers zero errors. $\square$

Barron–Cohen–Dahmen– DeVore prove in (Barron et al., 2008, Theorem 2.3, equation (2.31)) that exact OGA in a Hilbert space satisfies

$$\|f - f_N\|_H^2 \leq \|f - h\|_H^2 + \frac{4\|h\|_{L^1(\mathcal{D})}^2}{N}, \quad h \in L^1(\mathcal{D}). \quad (61)$$

Here $L^1(\mathcal{D})$ denotes the atomic variation space, not a spatial Lebesgue norm. The correspondence in our restarted setting is $f = Q_0 u^*$, $h = v_B$, $\|f - h\|_* \leq \sigma$, and atomic budget at most B_{entry} on the unrenormalized projected dictionary. The source pairing and reciprocal step above are the same classical mechanism. Theorem 5.10 retains the observed $\underline{\theta}_m^2/\ell_m^2$ along spaces selected by a nonlinear algorithm. Neither the approximate-source remainder nor the projection identity is claimed as a new general OGA principle. If $\underline{\theta}_m \geq \underline{\theta} > 0$, then $\ell_m \leq 1$ gives $W_j \geq j\underline{\theta}^2$. Stronger decay of the displayed bound requires independently controlled growth of W_j or stronger drop information.

For a direct comparison, Li–Siegel, Corollary 4.2 and equation (4.11), already use an approximate source h through an inequality of the form

$$\frac{1}{2}(\|r\|^2 - \|f - h\|^2) \leq \|h\|_{\mathcal{A}} |\langle r, g \rangle|,$$

which leads to the same innovation-weighted projection recursion. The present theorem specializes this mechanism to the restarted, unrenormalized projected dictionary and retains the observed θ_m^2/ℓ_m^2 weights generated by the nonlinear selection rule. It should therefore be read as a specialization to the computed sequence of approximation spaces, rather than as a new general approximate-source principle.

5.6 An exact quartic energy connection

For the one-dimensional pure or regularized quartic energy define

$$X = \{u \in W^{1,4}(0,1) : \int_0^1 u = 0\}, \quad \mathcal{E}_\epsilon(u) = \int_0^1 \left(\frac{\epsilon^2}{2} |u'|^2 + \frac{1}{4} |u|^4 \right) dx - \ell(u).$$

An additive constant in the density is immaterial. Assume $u^* \in X$ solves the weak equation and let $Y \subset X$ contain u^* and the spaces/dictionary of Section 5.5. Use

$$a_*(v, w) = \int_0^1 (\epsilon^2 + 3|(u^*)'|^2) v' w' dx.$$

For $\epsilon > 0$ this is positive definite on Y . For $\epsilon = 0$ assume explicitly that $\int |(u^*)'|^2 |v'|^2 > 0$ for every nonzero $v \in Y$; the sufficient condition $(u^*)' \neq 0$ almost everywhere ensures this on the zero-mean space. Finiteness follows from Holder's inequality. No full-space H^1 coercivity is inferred from positivity on Y .

For any $\widehat{G}_m \in \widehat{\Phi}_m$, define

$$D_m = \left\| \widehat{G}_m - z_m \right\|_*, \quad v_m = \widehat{G}_m' - (u^*)', \quad R_m = \int_0^1 ((u^*)' v_m^3 + \frac{1}{4} v_m^4) dx.$$

Proposition 5.11 (Quartic identity and comparison). *For every such state, with no exact-minimization requirement,*

$$\widehat{\delta}_m = \frac{1}{2}(r_m^2 + D_m^2) + R_m. \quad (62)$$

If $D_m \leq q_{\text{corr}} r_m$ and $|R_m| \leq \rho(r_m^2 + D_m^2)$ throughout a finite range, where $q_{\text{corr}} \geq 0$ and $0 \leq \rho < 1/2$, then

$$(\frac{1}{2} - \rho)r_m^2 \leq \widehat{\delta}_m \leq C_E r_m^2, \quad C_E = (\frac{1}{2} + \rho)(1 + q_{\text{corr}}^2). \quad (63)$$

If U_m is any source upper bound for r_m^2 , the unconditional additive estimate is $\widehat{\delta}_m \leq U_m/2 + D_m^2/2 + |R_m|$.

Proof. Set $a = (u^*)'$ and $v = v_m$. Subtracting the linear part of the density at a gives $\epsilon^2 v^2/2 + 3a^2 v^2/2 + av^3 + v^4/4$. The weak equation cancels the integrated linear part, including the load.

Also $\widehat{G}_m - z_m$ lies in the active space and $u^* - z_m$ is orthogonal to it, so $\left\| \widehat{G}_m - u^* \right\|_*^2 = D_m^2 + r_m^2$.

This proves the identity; the bounds follow by substitution. $\square$

The nonlinear correction and remainder are already evaluated at the computed state in (62). Adding a further optimization-error term to that identity would count the same effect twice. Quadrature error appears instead as a separately reported numerical defect. The constants q_{corr} and ρ computed using u^* are quantities evaluated using the manufactured solution; the identity alone does not provide solution-independent estimates of them. A small σ at the starting index need not be small relative to a terminal projection error.

5.7 Projection bounds under a power-type decrease condition

Retain the projections of Section 5.5 and set

$$x_m = r_m^2, \quad \gamma_m = \frac{|a_*(e_m, d_{m+1})|}{\ell_m}.$$

Then (60) is $x_{m+1} = x_m - \gamma_m^2$. When $S_m^{\text{fin}} > 0$, $\gamma_m = \theta_m S_m^{\text{fin}} / \ell_m$; otherwise $\gamma_m = 0$ by its direct definition. The word fractional in the following condition refers only to the power imposed on the residual, not to a fractional differential operator.

Theorem 5.12 (Explicit power-type finite-range bound). *Let $\alpha > 0$, $c_0 > 0$, and assume, at every step with $r_m > 0$ in $m_0 \leq m < m_0 + J$,*

$$\gamma_m^2 \geq c_0 r_m^{2+2/\alpha}. \quad (64)$$

For $r_{m_0} > 0$ and $0 \leq j \leq J$,

$$r_{m_0+j}^2 \leq \left(r_{m_0}^{-2/\alpha} + \frac{c_0 j}{\alpha} \right)^{-\alpha}. \quad (65)$$

If $r_{m_0} = 0$, every later projection error is zero. For the quartic models, if the relative conditions of Proposition 5.11 also hold, then

$$\widehat{\delta}_{m_0+j} \leq C_E \left(r_{m_0}^{-2/\alpha} + \frac{c_0 j}{\alpha} \right)^{-\alpha}. \quad (66)$$

Natural-distance and Sobolev bounds are obtained by taking respectively the one-half and one-quarter powers, with the constants of Theorem 4.6.

Proof. For $x_{m+1} > 0$, convexity gives

$$x_{m+1}^{-1/\alpha} - x_m^{-1/\alpha} \geq \frac{x_m - x_{m+1}}{\alpha x_m^{1+1/\alpha}} \geq \frac{c_0}{\alpha}.$$

Summation proves (65). If a residual reaches zero, nestedness gives the remaining cases. Substitute the projection bound into (63), then apply Theorem 4.6. $\square$

A sufficient factorization on positive-error steps is

$$S_m^{\text{fin}} \geq c_1 r_m^{2\nu}, \quad \theta_m / \ell_m \geq c_2, \quad \nu = \frac{1}{2} + \frac{1}{2\alpha}, \quad c_1, c_2 > 0. \quad (67)$$

Indeed $\gamma_m^2 \geq (c_1 c_2)^2 r_m^{4\nu}$ and $4\nu = 2 + 2/\alpha$. Thus $c_0 = (c_1 c_2)^2$ is admissible. The residual visibility factor and the selected-direction factor are separate requirements.

For a finite sequence with strictly positive drops, every prescribed $\alpha > 0$ admits

$$c_{\min}(\alpha; J) = \min_{\substack{m_0 \leq m < m_0 + J \\ x_m > 0}} \frac{x_m - x_{m+1}}{x_m^{1+1/\alpha}} > 0.$$

Using this value on the same sequence constructs an a posteriori envelope. It neither identifies a unique exponent nor independently guarantees a rate for CGA. Evaluating a fresh finite range with the constants fixed in advance supplies additional numerical evidence, but it still does not prove an asymptotic theorem. If any positive-error step has zero drop, no positive c_0 satisfies (64) on that whole finite range. For a fixed finite dictionary whose span does not contain the target, the positive best-approximation floor also prevents an indefinitely valid positive c_0 . Explicit bounds, their constants, and their finite-range lengths are therefore more informative than unqualified high-order notation.

6 CGA numerical experiments and finite-range estimates

6.1 Numerical setup, representative problems, and error quantities

All problems are posed on $\Omega = (0, 1)^d$, with $d = 1$ or 2 , and use manufactured profiles. The main suite uses smooth profiles; the final row is explicitly a low-order activation case $(p, k) = (5, 1)$, so its regularity statement is kept separate. The exact profiles, loads, homogeneous natural boundary data, feasible spaces, and casewise configurations are indexed in Supplement S1 and `manifest/experiment_index_v2.csv`. The exact solution is used to construct the load and to evaluate errors, but not in the greedy selection or coefficient optimization. Pure and regularized p -problems use a zero-mean representative, and their atoms are centered before normalization.

For a raw atom φ , let $P_\circ\varphi$ denote centering in the pure and regularized cases and the identity in the other cases. With Q_{tr} denoting the training quadrature, the experimental atom is $g_Q = P_\circ\varphi/\mathcal{N}_Q(P_\circ\varphi)$, where

$$\mathcal{N}_Q(g) = \begin{cases} [Q_{\text{tr}}(|g|^2 + |\nabla g|^2)]^{1/2}, & \text{linear and semilinear cases,} \\ [Q_{\text{tr}}(|\nabla g|^p)]^{1/p}, & \text{pure and regularized } p\text{-cases,} \\ [Q_{\text{tr}}(|g|^p + |\nabla g|^p)]^{1/p}, & \text{reaction } p\text{-case.} \end{cases}$$

This discrete homogeneous normalization is used for atom construction; it is distinct from the theoretically normalized continuous dictionary in Section 2 and from the nonlinear V -distance used to evaluate an approximation. Consequently, a discrete coefficient norm or training-quadrature normalization is not by itself an entropy or source bound in the analytical norm.

For coefficient optimization, residuals are evaluated in the stable coordinates supplied by the active feature basis. If a Euclidean coefficient gradient is used, the active Gram matrix is required to convert it to the dual norm on the active span; when the gradient is evaluated with the training quadrature, the corresponding quadrature error must also be included. Thus, an optimizer residual is not by itself a value of η_m^{stat} or $\varepsilon_m^{\text{opt}}$ in Theorem 3.5. The supplementary numerical data report the projected residual, the common target, the optimization status, and the energy decrease as separate quantities.

At each trial, the available atom with largest derivative score is tested, and all active outer coefficients are reoptimized when the innovation test succeeds. A retained step m is one that passes the rank, finite-value, stationarity, energy-decrease, and feasibility tests. Its active rank is N_m , while M denotes the last retained step reported for a run; other trials increase the trial counter but not m or N_m . In the present runs retained atoms are linearly independent, so the plotting abscissa $N = N_m = m$. The maximum attempted count specified by the configuration and all stopping reasons are recorded in the configuration and run summaries. The eight cases in Table 2 include a quadratic calibration, two semilinear energies, three $p = 4$ diffusion structures, a one-dimensional pure $p = 4$ problem, and a low-regularity $p = 5$ case.

For the main eight-case suite, the one-dimensional target width is 256 and the two-dimensional target width is 512. The separate quartic auxiliary numerical tests use $N = 8$ –32 for calibration and continue through $N = 64$. The one-dimensional pure $p = 4$ case reaches 141 retained atoms within 768 attempted selections and therefore has $N = 128$ as its last dyadic state. The CGA curves show the retained states over the reported range. Markers identify the powers of two used for local-order tables and are the only states joined by the display curves; the complete retained-state record is retained in the supplementary numerical data.

The quartic finite-trajectory calculation uses two finite index ranges. The constants and source quantities are calibrated only on the 25 retained states $N = 8, \dots, 32$; the subsequent states

Table 2: Representative CGA experiments and terminal retained widths.

Energy	d	p	k	Retained/target
linear reaction–diffusion, multifrequency	1	2	3	256/256
cubic semilinear, multifrequency	1	–	3	256/256
hyperbolic-sine semilinear	2	–	3	512/512
pure p -Laplacian, low-frequency profile	1	4	3	141/256
pure p -Laplacian	2	4	3	512/512
regularized p -Laplacian, $\varepsilon = 0.1$	2	4	3	512/512
reaction p -Laplacian	2	4	3	512/512
pure p -Laplacian	1	5	1	256/256

$N = 33, \dots, 64$ extend the same previously computed iterates and are not used to re-estimate those constants. Consequently, the number of transitions satisfying the normalized-innovation condition on $32 < N \leq 64$ measures whether the calibration remains informative immediately beyond its calibration range, whereas only the $8 \leq N \leq 32$ quantities enter the stated finite-range envelope. The two transition sets are reported separately below.

For the linear and semilinear cases we report the energy gap and

$$e_{H^1}(\hat{G}_N) = \frac{\|\hat{G}_N - u^*\|_{H^1}}{\|u^*\|_{H^1}}.$$

For a p -model the relative Sobolev error and relative natural distance are

$$e_{W^{1,p}}(\hat{G}_N) = \frac{\|\hat{G}_N - u^*\|_{W^{1,p}}}{\|u^*\|_{W^{1,p}}},$$

$$e_V(\hat{G}_N) = \frac{d_{V,\bullet}(\hat{G}_N, u^*)}{d_{V,\bullet}(0, u^*)}.$$

The benchmark solutions make these denominators nonzero. The energy gap is the absolute quantity $\hat{\delta}_N = \mathcal{E}(\hat{G}_N) - \mathcal{E}(u^*)$. The dyadic local order is

$$\text{ord}_N(e) = \log_2 \left(\frac{e(N/2)}{e(N)} \right).$$

These three quantities are kept separate in every figure, table, and subsequent comparison.

6.2 Conditional reference rates on finite index ranges

For ReLU^k atoms, the Hilbert-space approximation exponent associated with the smooth-ridge entropy estimate is

$$\beta_H = \frac{1}{2} + \frac{2(k-1)+1}{2d}.$$

The dashed segments in the CGA convergence figures show the conditional powers $N^{-2\beta_H}$ for the energy gap and $N^{-\beta_H}$ for either the H^1 error or the natural V -distance. For a standard p -structure energy, the corresponding comparison power for the $W^{1,p}$ error is $N^{-2\beta_H/p}$, as follows from the energy-to-metric conversion in Corollary 4.7. The exponent is fixed by the stated theory and is not fitted to the curve. The vertical position is chosen once from the last up to three positive dyadic states by setting $\log C = \text{median}_i(\log e_i + s \log N_i)$, where $e_i > 0$ are those terminal values and s

is the prescribed exponent. This minimizes the sum of absolute log deviations for the reference segment CN^{-s} . The segment starts at $N = 8$ in one dimension and at $N = 16$ in two dimensions and extends to the largest retained width shown in the same panel. The displayed index range is not a claim that the continuous coverage, source, and transfer assumptions have been verified on that interval.

Table 3: Exponents used only for the conditional reference segments in the CGA figures.

(k, d, p)	β_H	Energy rate	H^1/V rate	$W^{1,p}$ rate
(1, 1, 5)	1	2	1	2/5
(3, 1, 4)	3	6	3	3/2
(3, 2, 4)	7/4	7/2	7/4	7/8
(3, 1, --)	3	6	3	–
(3, 2, --)	7/4	7/2	7/4	–

6.3 Linear and cubic calibration

Figure 1 shows the retained dyadic states for the linear and cubic multifrequency problems. Both curves decrease rapidly after about 32 atoms. Between $N = 128$ and 256, the linear energy and H^1 orders are 6.083 and 3.042, whereas the cubic orders are 6.896 and 3.448; the corresponding terminal relative H^1 errors are 1.247×10^{-4} and 1.215×10^{-4} . These are measured local orders on one resolved interval, not estimates of an asymptotic exponent.

Table 4: Dyadic errors for Linear reaction–diffusion, d=1.

N	Energy gap	Order	Relative H^1 error	Order
32	4.277×10^{-1}	–	1.323×10^{-1}	–
64	3.983×10^{-3}	6.747	1.277×10^{-2}	3.373
128	2.577×10^{-5}	7.272	1.027×10^{-3}	3.636
256	3.801×10^{-7}	6.083	1.247×10^{-4}	3.042

Table 5: Dyadic errors for Cubic semilinear problem, d=1.

N	Energy gap	Order	Relative H^1 error	Order
32	3.752×10^{-1}	–	1.239×10^{-1}	–
64	2.238×10^{-3}	7.389	9.569×10^{-3}	3.694
128	4.301×10^{-5}	5.702	1.326×10^{-3}	2.851
256	3.610×10^{-7}	6.896	1.215×10^{-4}	3.448

The two-dimensional sinh case is reported in the supplementary figures. Its relative H^1 error decreases from 4.074×10^{-3} at $N = 256$ to 1.990×10^{-3} at $N = 512$, while the local order decreases from 1.793 on $128 \rightarrow 256$ to 1.034 on $256 \rightarrow 512$. The final interval therefore exhibits slower decay than the preceding interval.

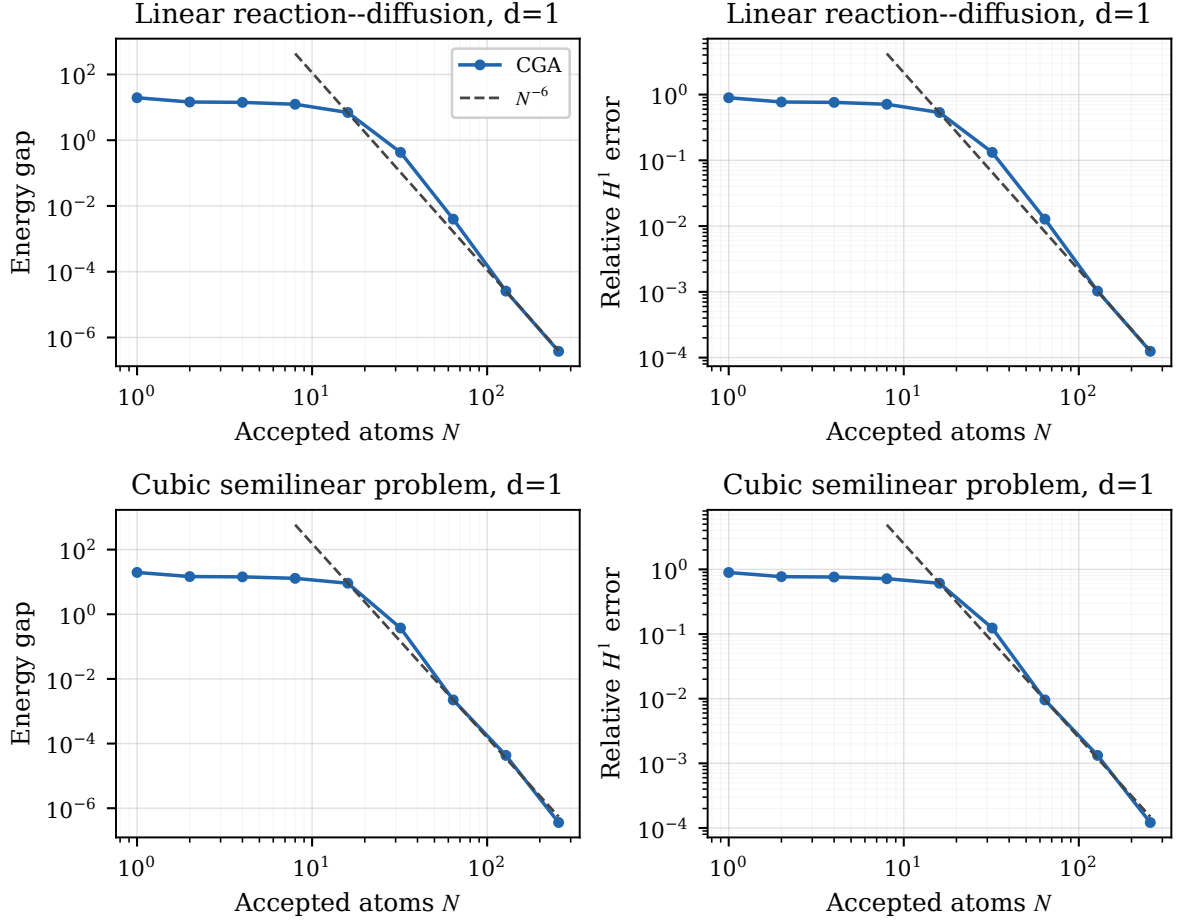


Figure 1: CGA values at retained powers of two for the linear and cubic problems. Each dashed segment is the corresponding metric-specific conditional reference rate of Table 3, restricted to the displayed index range.

6.4 Numerical results for the pure and regularized $p = 4$ models

Figure 2 presents the one- and two-dimensional pure $p = 4$ cases using energy, relative $W^{1,4}$ error, and relative V -distance. The one-dimensional case reaches relative errors 1.579×10^{-5} in $W^{1,4}$ and 1.680×10^{-6} in the V -distance at its last dyadic state $N = 128$. For the two-dimensional pure $p = 4$ problem, the interval $128 \rightarrow 256$ has energy and V -orders 3.410 and 1.703. On the final interval the V -distance continues to decrease, but the relative $W^{1,4}$ error increases from 2.494×10^{-3} to 3.601×10^{-3} . Thus the terminal Sobolev-error behavior is nonmonotone even though the V -distance decreases.

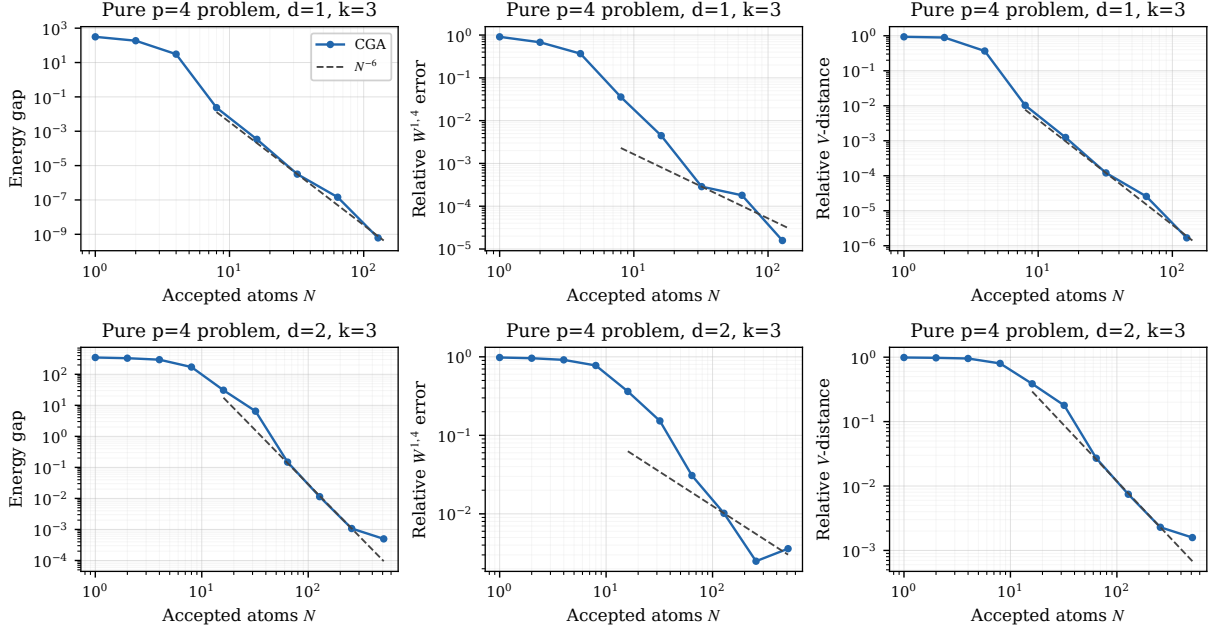


Figure 2: CGA values at retained powers of two for the representative one- and two-dimensional pure $p = 4$ problems. The three columns show energy gap, relative $W^{1,4}$ error, and relative V -distance. Dashed segments use the metric-specific conditional exponents in Table 3.

Table 6: Dyadic errors for Pure $p=4$ problem, $d=1$, $k=3$.

N	Energy gap	Order	Relative $W^{1,4}$	Order	Relative V	Order
8	2.363×10^{-2}	—	3.567×10^{-2}	—	1.028×10^{-2}	—
16	3.372×10^{-4}	6.131	4.460×10^{-3}	2.999	1.241×10^{-3}	3.051
32	3.180×10^{-6}	6.728	2.869×10^{-4}	3.959	1.205×10^{-4}	3.364
64	1.441×10^{-7}	4.464	1.800×10^{-4}	0.672	2.565×10^{-5}	2.232
128	6.187×10^{-10}	7.864	1.579×10^{-5}	3.511	1.680×10^{-6}	3.932

To examine the transition from uniformly elliptic regularizations to the pure model on one common finite index range, Figure 3 compares $\varepsilon = 0, 10^{-2}, 10^{-1}, 1$ for $p = 4, d = 1, k = 3$. At $N = 32$, the four relative $W^{1,4}$ errors lie between 3.209×10^{-4} and 3.261×10^{-4} , while the relative V -distances lie between 9.518×10^{-5} and 1.198×10^{-4} . The spread of these values indicates weak dependence on the regularization over this parameter range, and it does not establish coercivity of the unregularized Hessian on the full energy space.

The regularized and reaction $p = 4$ trajectories are shown in the supplementary figures. As in

Table 7: Dyadic errors for Pure $p=4$ problem, $d=2$, $k=3$.

N	Energy gap	Order	Relative $W^{1,4}$	Order	Relative V	Order
16	3.070×10^1	—	3.624×10^{-1}	—	3.862×10^{-1}	—
32	6.476×10^0	2.245	1.528×10^{-1}	1.247	1.786×10^{-1}	1.113
64	1.479×10^{-1}	5.452	3.078×10^{-2}	2.311	2.701×10^{-2}	2.725
128	1.140×10^{-2}	3.697	1.017×10^{-2}	1.598	7.452×10^{-3}	1.858
256	1.073×10^{-3}	3.410	2.494×10^{-3}	2.028	2.288×10^{-3}	1.703
512	4.975×10^{-4}	1.109	3.601×10^{-3}	-0.530	1.584×10^{-3}	0.531

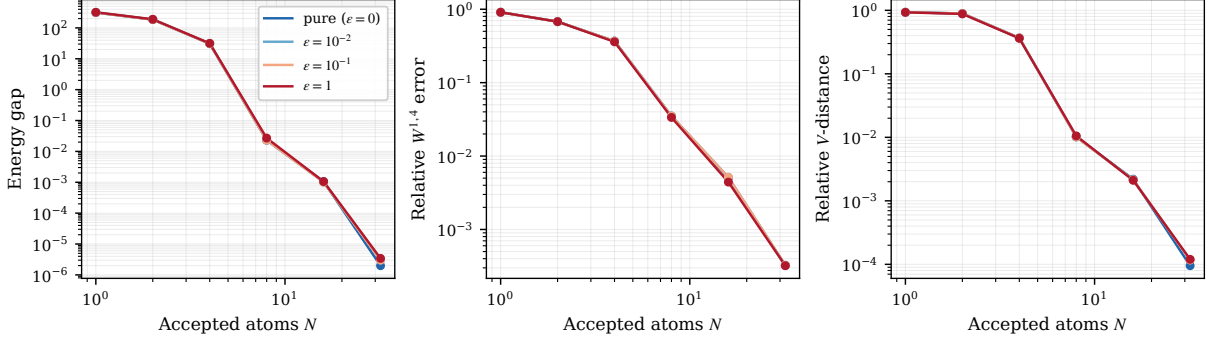


Figure 3: Pure and regularized $p = 4, d = 1, k = 3$ CGA trajectories for four values of ε . Only dyadic retained states are connected. All runs use the same target width and evaluator.

the pure two-dimensional case, their final $W^{1,4}$ errors increase while their V -distances continue to decrease. The same nonmonotone terminal behavior therefore appears in all three $p = 4$ structures.

6.5 Finite-trajectory bounds and fractional innovation for the quartic models

We evaluate the estimates of Section 5 on the one-dimensional pure and regularized $p = 4, k = 3$ trajectories, with $u^*(x) = \cos(2\pi x)$ and $\varepsilon = 0$ or 0.1 (cases `base_pure` and `epsilon_01` in Supplement S1). These runs use a larger pool and a smaller accepted-state budget than the eight cases of Section 6. The inner product held fixed during the estimates is

$$a_*(v, w) = \int_0^1 (\varepsilon^2 + 3|(u^*)'|^2) v' w' dx.$$

The calibration range contains the 25 retained states $N = 8, \dots, 32$ and its 24 transitions $N \rightarrow N+1$, $N = 8, \dots, 31$. The subsequent finite range contains the 32 transitions $N \rightarrow N+1$, $N = 32, \dots, 63$, generated by extending the same previously computed iterates to $N = 64$. The two transition sets are kept separate: all source, comparison, and fractional-innovation constants are estimated on the calibration range only, while the continuation is used only as a auxiliary numerical test. At the starting index, the fixed auxiliary dictionary comprises 463 valid candidates and the first 128 valid reference atoms in the order in which they were generated, drawn with a separate seed. They need not be linearly independent. Symmetry and the frozen normalization follow the theory, and Supplement S2 specifies the ordering and the filtering. The reference atoms enter the estimates only. Source coefficients are constructed from the starting-index residual and are not adjusted to fit later errors.

The full active-space projections use untruncated QR on a common breakpoint-segmented Gauss rule of order 20, cross-checked at order 16. The active linear systems have condition numbers

between 7.7×10^1 and 4.1×10^3 . The maximum defect in Equation (62) is below 8.7×10^{-17} , and the projection-drop identity has residual below 1.5×10^{-12} . These are floating-point calculations, not interval enclosures of the exact integrals or projections.

Source bound and Hessian-transfer conditions. Table 8 collects the source data, the quartic comparison constants, and the initial-value bounds of Theorem 5.10, evaluated with $\underline{\theta}_m = \theta_m$, the observed selection ratio. In the table, $B = B_{\text{entry}}$ and $q = q_{\text{corr}}$. The source energy bounds at $N = 32$ are 2.23×10^{-2} and 2.11×10^{-2} , compared with energy gaps 1.99×10^{-6} and 3.01×10^{-6} . Thus the source remainder does not make this bound sharp; the accumulated-innovation contribution remains dominant. For both models, σ^2/r_8^2 is approximately 6.21×10^{-5} , whereas σ^2/r_{32}^2 is 0.751 for the pure model and 0.495 for the regularized model.

Table 8: Source and quartic comparison quantities on all retained states $N = 8\text{--}32$. The source is fixed at the starting index; q and ρ are calibration-range maxima on $N = 8\text{--}32$. Values are floating-point estimates.

Quantity	Pure $p = 4$	$\varepsilon = 0.1$
Source budget B	4.361×10^1	4.362×10^1
Source remainder σ	1.726×10^{-3}	1.728×10^{-3}
$\mathcal{W}_{24}$	1.912×10^4	2.913×10^4
$q = \max D_N/r_N$	4.733×10^{-2}	4.715×10^{-2}
$\rho = \max R_N /(r_N^2 + D_N^2)$	2.006×10^{-2}	1.997×10^{-2}
Computed r_{32}^2	3.969×10^{-6}	6.025×10^{-6}
Initial-value source bound for r_{32}^2	4.280×10^{-2}	4.057×10^{-2}
Computed energy gap at $N = 32$	1.985×10^{-6}	3.013×10^{-6}
Initial-value source energy bound	2.231×10^{-2}	2.114×10^{-2}

For comparison with the finite-range discussion, the minimum active-space generalized Hessian eigenvalues relative to the H^1 Gram matrix are 0.1328 and 0.1428, and the minimum finite-reference-pool selection ratios are 0.9876 and 0.9911. Nevertheless, the sufficient lower bound of Equation (55) is negative at all 24 transitions in both models. Positive active-space spectra and finite-reference-set ratios therefore do not establish the continuous relative-score condition. The selected directions remain valid for the finite numerical experiment, but the transfer proposition cannot be invoked from these quantities alone. The direct projection estimates below use the selected directions themselves and do not require that sufficient lower bound to be positive. The decomposition in Table 9 shows one representative transition. The Hessian-transfer lower bound is negative even though the corresponding transition is retained; the score-quadrature and active-optimization terms are negligible at this step relative to the nonlinear-to-frozen defect.

Table 9: One-step decomposition of the error terms at the first calibration transition. The negative selection lower bound indicates that the nonlinear-to-frozen defect is larger than the available frozen decrease, although the corresponding transition is retained by the numerical procedure.

Quantity	Pure $p = 4, 8 \rightarrow 9$	Regularized $\varepsilon = 0.1, 8 \rightarrow 9$
Frozen decrease	1.8403×10^{-2}	1.8175×10^{-2}
Nonlinear-to-frozen defect	6.8347×10^{-3}	6.8108×10^{-3}
Score quadrature defect	9.73×10^{-13}	9.95×10^{-13}
Active optimization defect	4.36×10^{-7}	4.48×10^{-7}
Total score defect	6.8351×10^{-3}	6.8112×10^{-3}
Selection lower bound	-6.3599×10^{-2}	-6.2742×10^{-2}

Fractional-innovation calculation. For the benchmark $\alpha = 6$, fixed before evaluating the normalized drops, set

$$x_N = r_N^2, \quad \gamma_N = \sqrt{x_N - x_{N+1}}, \quad c_N = \frac{x_N - x_{N+1}}{x_N^{7/6}}, \quad N = 8, \dots, 31. \quad (68)$$

The minimum over all transitions is 1.6904×10^{-2} for the pure model and 1.0912×10^{-2} for $\varepsilon = 0.1$; both occur at $25 \rightarrow 26$. Changing the quadrature order changes these stepwise ratios by less than 5.6×10^{-12} relatively. Together with the computed q and ρ , these values permit a numerical evaluation of Equations (65) and (66). Writing $j = N - 8$, the resulting finite-range comparison powers are

$$\hat{\delta}_{8+j} = O(j^{-6}), \quad d_{V,\bullet}(\hat{G}_{8+j}, u^*) = O(j^{-3}), \quad \|\hat{G}_{8+j} - u^*\|_{W^{1,4}} = O(j^{-3/2}), \quad 1 \leq j \leq 24.$$

The explicit bound in Equation (65) also includes the starting state $j = 0$. These powers describe a conditional envelope on the tested finite range, not a limit as $N \rightarrow \infty$.

Table 10: Fractional-innovation quantities for $\alpha = 6$, computed from every transition $N \rightarrow N + 1$, $N = 8, \dots, 31$. The fitted exponent describes the innovation score and is not used to choose the benchmark exponent or its bound.

Quantity	Pure $p = 4$	$\varepsilon = 0.1$
$c_{\min}$ over all 24 transitions	1.690×10^{-2}	1.091×10^{-2}
Transition attaining $c_{\min}$	$25 \rightarrow 26$	$25 \rightarrow 26$
Fitted slope ν of $\log \gamma_N$ versus $\log x_N$	0.5866	0.5898
$1/(2\nu - 1)$ from the fit	5.77	5.57
Fractional bound for r_{32}^2 , $\alpha = 6$	3.774×10^{-2}	4.110×10^{-2}
Fractional bound / computed r_{32}^2	9.510×10^3	6.821×10^3
Fractional energy bound at $N = 32$	1.967×10^{-2}	2.142×10^{-2}

A least-squares fit of $\log \gamma_N$ against $\log x_N$, using all 24 transitions, gives slopes $\nu = 0.5866$ and 0.5898 . The relation $\alpha = 1/(2\nu - 1)$ associates these fitted innovation powers with 5.77 and 5.57, respectively. This fit concerns $\gamma_N = \theta_N S_N / \ell_N$, not S_N alone; it does not independently establish either factor in Equation (67).

The worst-step constants still yield conservative terminal projection bounds: 3.77×10^{-2} and 4.11×10^{-2} , approximately 9.5×10^3 and 6.8×10^3 times the computed projection errors. The fractional bound is slightly smaller than the source bound for the pure model and slightly larger for the regularized model. Positive drops on a finite sequence also give a positive minimum normalized ratio for any fixed exponent. The computation establishes the stated numerical envelope and its constants, but does not uniquely identify sixth-order decay or verify an asymptotic mechanism. Such a conclusion would require bounds uniform over increasing index ranges.

Continuation beyond the calibration range. To test whether the calibrated constants remain informative immediately beyond the range on which they were fixed, we continued the two trajectories from $N = 32$ to $N = 64$, keeping the same candidate pools, previously computed iterates, and source coefficients and drawing no further random atoms. The comparison exponent and all constants were kept at their $N = 8$ –32 values. Most transitions in this subsequent finite range satisfy the normalized-innovation sufficient condition with those constants. At quadrature order 20, it holds for 27 of the 32 continuation transitions in the pure model $27/32 = 84.4\%$ and 28 of

32 in the regularized model $28/32 = 87.5\%$; only five and four transitions, respectively, fall below the prescribed threshold. These exceptions concern the sufficient condition, not the validity of the theoretical implication. In both models, the first such transition is $N = 33 \rightarrow 34$, and the minimum ratio c_N/c_0 over the continuation is 0.156 and 0.187, respectively; the terminal squared projection errors are 4.770×10^{-7} and 4.626×10^{-7} . The order-16 cross-check changes these reported values only in the displayed last digits. The continuation therefore shows that the calibrated condition remains satisfied for most transitions, together with a substantial error decrease and agreement with the algebraic identities on the subsequent states. The few exceptions prevent applying the same fixed-constant theorem over the entire continuation range. The stated finite-range envelope therefore retains its calibration range $N = 8\text{--}32$, while the continuation provides supporting stepwise numerical evidence without establishing an asymptotic rate.

Table 11: Continuation from $N = 32$ to $N = 64$ (32 transitions) with constants fixed on $N = 8\text{--}32$. The number of transitions satisfying the normalized-innovation sufficient condition and the first transition not satisfying it are reported.

Model	New steps	Number satisfying	First not satisfying N	$\min c_N/c_0$	r_{64}^2
Pure $p = 4$	32	27	33	0.156	4.770×10^{-7}
$\varepsilon = 0.1$	32	28	33	0.187	4.626×10^{-7}

7 Equal-DOF baseline comparisons

7.1 Comparison setup and scope

Five cases use a common evaluator: the one-dimensional linear problem, the one-dimensional cubic problem, the two-dimensional sinh problem, the one-dimensional pure $p = 4$ problem, and the two-dimensional pure $p = 4$ problem. The comparison variable is the number of independent global coefficients: retained atoms for CGA, ridge features held fixed for RFM, and independent finite-element coefficients after the constant mode is removed where required. Equal degrees of freedom (DOF) do not imply equal runtime, memory, score evaluations, or nonlinear solver work, and no cross-method timing conclusion is drawn from runs performed in different environments. The supplementary numerical data at <https://github.com/RonzolYu/cga4PDE> give the prescribed realization count, the available converged realizations, and the solver status for each width.

FEM uses continuous Lagrange elements on uniform one-dimensional meshes or structured triangular two-dimensional meshes. P1 and P3 provide low- and high-order mesh-based references in the main figures. P2 results are retained in the supplementary material. P3 is the most accurate among the tested uniform-mesh FEM variants at each largest common coefficient count reported below. The linear and cubic FEM curves are close at high resolution, which is consistent with the common manufactured profile and an H^1 error dominated by the finite-element approximation.

For each case, the RFM uses a nested family of normalized ReLU³ ridges and fits the outer coefficients. Ten random-feature realizations are prescribed for each problem. Curves show the median and interquartile range over available converged realizations at each width, and the supplementary numerical data give the prescribed count together with solver status. All ten realizations reach the displayed ranges for the linear, cubic, sinh, and one-dimensional pure $p = 4$ problems. In the two-dimensional pure $p = 4$ problem, seven realizations are available at $N = 128$ and four are available at $N = 256$; consequently its statistical curve ends at 256 and its terminal band is based on four available realizations. No distributional statement is made at $N = 512$.

The CGA and RFM comparison curves join the computed powers of two listed in the numerical setup and the terminal retained state needed to define the largest common comparison width, while FEM uses its own mesh degrees of freedom. The figures contain no common-grid interpolants, theoretical reference lines, or convergence-order-versus-DOF curves. The reported panel range ends at the terminal CGA width fixed by the stopping rule, so that endpoint is determined by the stopping rule and the available method states rather than by the plotted error values. Each method is shown only over the states it actually reached. The equal-DOF tables use linear interpolation of $\log(\text{error})$ against $\log(\text{DOF})$ between adjacent computed values when an exact coefficient count is unavailable; no extrapolation is used.

Table 12: Training and evaluation parameters for the equal-DOF comparison.

Component	Setting
RFM	Nested ReLU ³ prefixes; the discrete homogeneous normalization defined in Section 6; centered for pure p ; ten prespecified random seeds, with prescribed and available-realization counts reported.
FEM	Continuous P1/P2/P3; structured meshes; assembly order 10, increased to 80 for the one-dimensional multifrequency cases; relative/absolute Newton tolerances $10^{-11}/10^{-12}$.
Training quadrature	One dimension: 256 cells, order 6; two dimensions: 63×63 cells, tensor-product order 3.
Evaluation quadrature	One dimension: 2048 cells, order 8; two dimensions: 64×64 cell partition, order 3.
Independent refinement	One dimension: 3072 cells, order 8; two dimensions: 80×80 cell partition, order 3.
Common DOFs	8, 16, 32, 64, 128, 256, with 512 used only when every method required by the comparison is bracketed by computed states.

7.2 Unified evaluator, retained states, and fixed denominators

The comparison states are re-evaluated after training by a frozen evaluator that is separate from the quadrature used to construct the features or fit their coefficients. The main evaluation rule uses the quadrature listed in Table 12; an independent refinement is used for the independently evaluated quantities. Thus, a solver-specific status is not interpreted as an independent accuracy bound, and a state is not removed merely because an independently evaluated quantity is unavailable. The set of retained states and the denominator conventions are summarized in Tables 13 and 14.

The common evaluation retains 617 stored states, comprising 612 newly generated states and five archived terminal states. All 617 satisfy their solver-specific acceptance criterion and the independent coefficient-count check. The stricter signed energy/metric evaluation admits 611 states, while the common H^1 -dual residual threshold admits 394; these are different filters and are reported with their own denominators. The independent two-dimensional Bregman calculation is an energy-integration evaluation on 138 states, not a replacement for the residual or optimizer checks.

For random-feature comparisons, the prescribed denominator is ten seeds at every configured width. The displayed median and interquartile range are computed only from evaluated states that pass the metric-specific checks, whereas missing rank-budget states remain failures in the 10-seed denominator. In particular, the three C3 states missing at $N = 256$, the one C5 state missing at $N = 128$, and the two C5 states missing at $N = 256$ are not dropped before forming success counts. All other displayed RFM widths have 10/10 evaluated states; their strict residual counts can still be smaller because residual qualification is a separate filter.

Table 13: Common re-evaluation set and prescribed denominators for the retained comparison states.

Quantity or convention	Count or convention
Retained states in the common evaluation	617 total: 612 newly generated; 5 archived terminal
Acceptance according to the solver-specific criterion	(617/617); this records the solver-specific acceptance rule only
Independent coefficient-count check	(617/617)
Independent signed energy/metric evaluation	(611/617); six states remain excluded from the corresponding strict energy summary
Common H1-dual residual, threshold 2e-6	(394/617); this is reported separately from solver-specific acceptance
Independent two-dimensional Bregman-energy evaluation	(137/138); this tests energy-integration stability and does not establish optimization accuracy

Table 14: Prescribed random-feature denominators at widths where states are missing or additional independent filtering is applied. A missing state remains in the denominator.

Case/width	Plan	Eval.	Missing	Energy	Strict H1 residual
C3, (N=256)	10	7	3	(7/10)	(0/10)
C5, (N=128)	10	9	1	(9/10)	(0/10)
C5, (N=256)	10	8	2	(7/10)	(0/10)

7.3 Relative Sobolev errors

For the linear, cubic, and sinh problems, Figures 4 and 5 report the relative H^1 error. For the two pure $p = 4$ problems, the corresponding panels report the relative $W^{1,4}$ error. These are Sobolev errors; the natural V -distance is reported separately below.

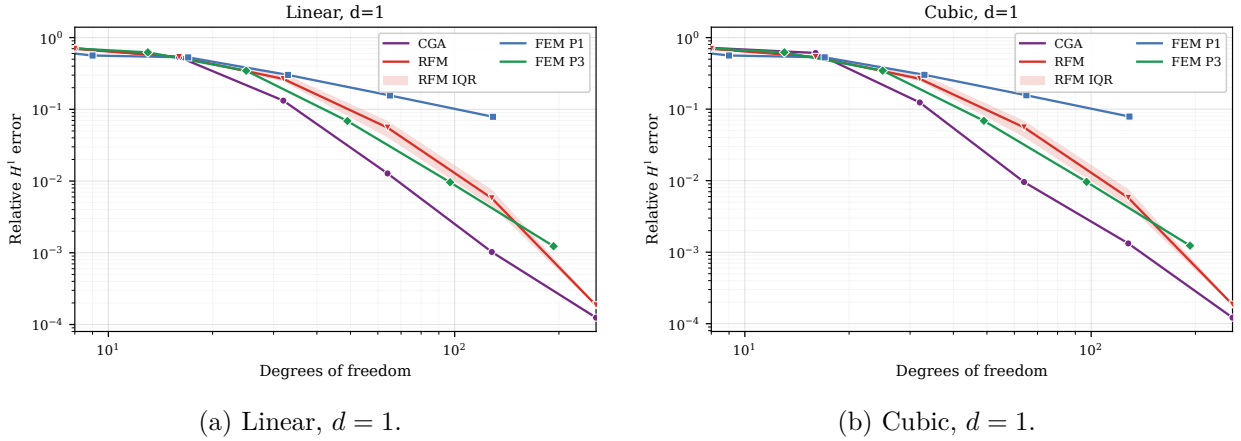


Figure 4: Relative H^1 error versus effective degrees of freedom for the one-dimensional multifrequency problems. RFM curves show medians and interquartile bands over successful realizations from ten prespecified runs. All markers are computed states.

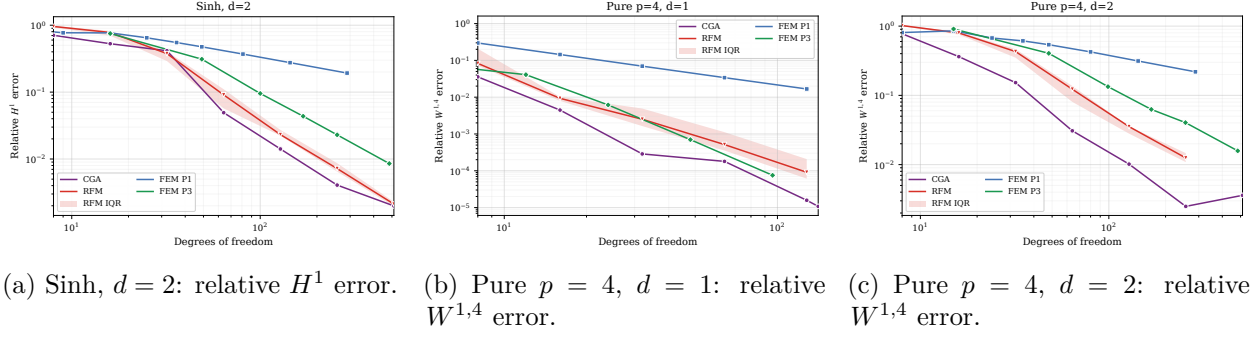


Figure 5: Relative Sobolev errors for the two-dimensional sinh problem and the pure $p = 4$ problems. The two-dimensional pure $p = 4$ RFM band is formed from the available realizations; its statistics at $N = 128$ and 256 use seven and four available realizations, respectively.

At the largest common endpoint supported by the configured RFM widths and bracketing FEM states, CGA has the smallest relative Sobolev error in the linear, cubic, sinh, and one-dimensional pure $p = 4$ problems, for which all ten prescribed RFM solves succeeded. The RFM-median/CGA ratios are 1.49, 1.52, 1.77, and 5.76, and the P3/CGA ratios are 4.24, 4.35, 5.64, and 1.88. In the two-dimensional pure $p = 4$ problem the successful-run RFM median and the P3 value are also above the CGA value, by factors 5.01 and 16.12, respectively, but the RFM statistic is based on only four of ten prescribed solves. These are endpoint comparisons at equal coefficient count. The curves cross at smaller widths in some cases, so the data do not support uniform dominance at every DOF.

Table 15: Relative Sobolev error at the largest common coefficient count reported for each problem. RFM statistics use successful realizations from ten prescribed runs and are reported as median [Q1,Q3]. The first four problems have ten successful runs; the two-dimensional pure $p = 4$ row is conditional on four successful runs and is not included in an aggregate ranking. Ratios larger than one mean that the corresponding comparison error is larger than the CGA error.

Problem	DOF	CGA		RFM median [Q1,Q3]	Success	RFM/CGA	FEM P3/CGA
Linear, d=1	256	1.247×10^{-4}	1.853×10^{-4}	$[1.779 \times 10^{-4}, 1.913 \times 10^{-4}]$	10/10	1.49	4.24
Cubic, d=1	256	1.215×10^{-4}	1.853×10^{-4}	$[1.779 \times 10^{-4}, 1.913 \times 10^{-4}]$	10/10	1.52	4.35
Sinh, d=2	256	4.074×10^{-3}	7.195×10^{-3}	$[6.472 \times 10^{-3}, 8.663 \times 10^{-3}]$	10/10	1.77	5.64
Pure p=4, d=1	128	1.579×10^{-5}	9.103×10^{-5}	$[6.108 \times 10^{-5}, 2.077 \times 10^{-4}]$	10/10	5.76	1.88
Pure p=4, d=2	256	2.494×10^{-3}	1.248×10^{-2}	$[1.087 \times 10^{-2}, 1.465 \times 10^{-2}]$	4/10	5.01	16.12

7.4 Natural V -distance for the p -models

Figure 6 repeats the two pure $p = 4$ comparisons with the relative V -distance defined in Section 6. Keeping this geometry separate from the $W^{1,4}$ error is important because the two quantities need not have the same terminal behavior. The figure uses the same computed-state and RFM uncertainty conventions as the Sobolev panels.

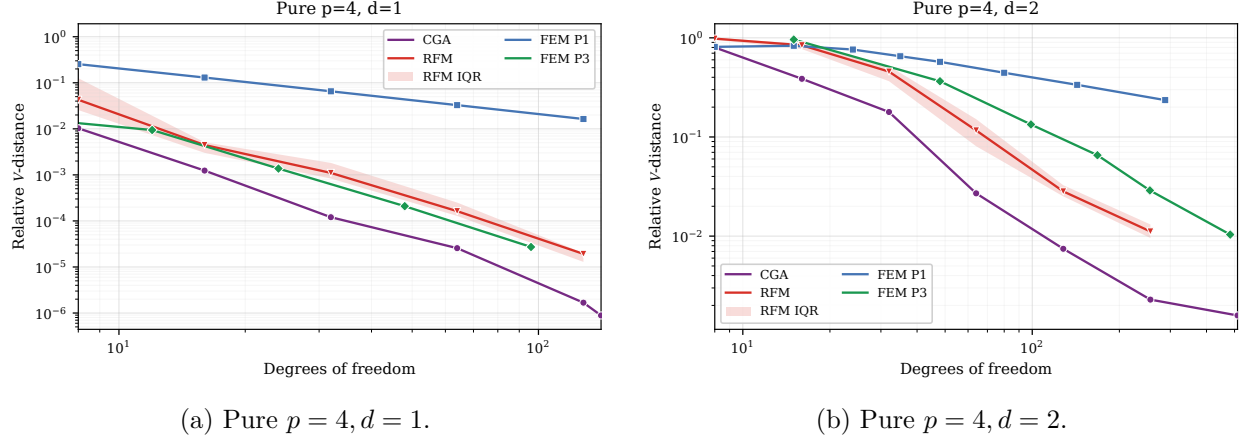


Figure 6: Relative V -distance versus effective degrees of freedom. No theoretical reference line or common-grid interpolation is drawn.

7.5 Interpretation limits

The equal-DOF results compare finite-pool CGA with the specified FEM and RFM implementations. They do not establish a runtime ordering, do not compare CGA with every randomized or greedy dictionary method, and do not turn endpoint ratios into asymptotic complexity estimates. The two-dimensional pure $p = 4$ RFM summaries use the four available realizations at the terminal width and are not used as a complete ten-run statistical comparison. Supplementary energy-gap comparisons are more sensitive to quadrature because they subtract two separately integrated energies. The supplementary numerical data specify the prescribed and converged realization counts, interpolation brackets, and computed coefficient counts.

8 Discussion and conclusion

The atomic source condition and directional smoothness yield an $O(m^{-1})$ energy bound for the ideal continuous-dictionary CGA. For finite-pool CGA, continuous-dictionary coverage enters multiplicatively, whereas score evaluation, radial stationarity, and coefficient optimization introduce distinct additive errors. The one-step inequality specifies the relative accuracy conditions under which the same energy order is retained.

For standard pure, regularized, and reaction p -energies with $p \geq 2$, the energy gap is equivalent to the squared natural distance and controls the p -th power of the Sobolev error. An energy exponent α therefore gives natural-distance and Sobolev exponents $\alpha/2$ and α/p . This conclusion uses the pointwise structure of the energy density; a general convex density with p -growth need not satisfy the same equivalence.

The sharper estimates in Section 5 require additional assumptions. Residual comparison relates the dictionary seminorm to the full dual norm. Local Hessian transfer requires coercivity, a dictionary norm comparison, relative score resolution, and source and entropy bounds after restart. The finite-trajectory estimates use the projections onto the computed spaces, together with either an approximate source representation or a fractional lower bound on the normalized innovations. The quartic energy decomposition relates both projection estimates to nonlinear energy errors.

The concrete semilinear family in Section 5.3 makes the nonlinear comparison explicit. Its C_{emb} , L_λ , C_λ , and B_λ constants give a computable lower bound Δ_λ with a C_λ^2 factor. The conditional

rate also requires a quantitative entropy bound for the same projected, unrenormalized restarted dictionary and invariance of the Hessian ball. These conditions remain to be verified uniformly before an unconditional rate follows.

The one-dimensional pure and regularized quartic calculations evaluate these quantities at every retained state on $N = 8\text{--}32$ and continue the same prefixes through $N = 64$ with the constants held fixed. Active-space Hessian spectra are positive, but the sufficient selection lower bounds are negative throughout this index range. The continuous Hessian-transfer hypotheses are therefore unverified. The direct projection and energy identities agree to floating-point accuracy under two quadrature rules. Both the source and fractional-innovation bounds cover the computed errors, although the terminal bounds remain several thousand times larger. With the prescribed exponent $\alpha = 6$, the latter yield finite-range comparison powers 6, 3, and $3/2$ for energy, natural distance, and $W^{1,4}$ error. The fitted innovation powers are descriptive: positivity on a finite trajectory neither determines a unique decay exponent nor gives constants uniform over increasing widths. In the continuation, the normalized-innovation sufficient condition holds on most transitions: 27 of 32 for the pure model and 28 of 32 for the regularized model. The few exceptions preclude a uniform application of the frozen-constant bound across the extended index range, so the continuation remains stepwise numerical evidence.

At the largest common coefficient count, CGA has the smallest relative Sobolev error in the four comparisons after filtering to the reported converged RFM realizations. The same ordering holds in the two-dimensional pure $p = 4$ example, where only four of ten prescribed RFM realizations reach the terminal width. These comparisons concern the specified implementations and coefficient counts; they establish neither an ordering at every width nor a runtime advantage.

Further work should either establish the three sufficient conditions for a specified nonzero λ range, or provide an independently supported error-decomposition tolerance rule. More examples alone would not resolve these gaps. Reliable numerical enclosures and comparisons with a common computational-work measure would provide additional evidence beyond the finite-range comparisons reported here.

A Proof of the one-dimensional source estimate

We give the details of Proposition 2.3 because it provides a concrete source class without appealing to a multivariate variation-space theorem. The Sobolev representative of $v \in W^{k+1,1}(0,1)$ satisfies the integral remainder identity

$$v(x) = \sum_{j=0}^k \frac{v^{(j)}(0)}{j!} x^j + \frac{1}{k!} \int_0^1 v^{(k+1)}(t) (x-t)_+^k dt. \quad (69)$$

By hypothesis, the polynomial part is a finite linear combination of fully active ridges. The coefficients are bounded by a constant times $\sum_{j=0}^k |v^{(j)}(0)|$, where the constant accounts for the conditioning of the fixed basis of $\mathbb{P}_k$, the space of polynomials of degree at most k . It also depends on the chosen Sobolev norm and on the fixed exponent p ; rescaling the interval would change it.

For the second term, the map

$$[0,1] \ni t \longmapsto (\cdot - t)_+^k \in W^{1,p}(0,1)$$

is continuous for $k \geq 1$. Indeed, both the function and its first weak derivative depend continuously on the translation parameter in the corresponding L^p spaces. Hence the integral in Equation (69) is a Bochner integral. Approximate $v^{(k+1)}(t) dt/k!$ by signed simple measures. The associated finite

sums of truncated powers converge in $W^{1,p}$, and their total coefficient variation is bounded by $\|v^{(k+1)}\|_{L^1}/k!$. Passing to the atomic closure and including the upper normalization factor proves Proposition 2.3.

The argument also shows why $k \geq 1$ is used when the ambient space is $W^{1,p}$: a discontinuous step activation does not have the required translation continuity in that norm. It shows in addition why the multivariate case is not obtained by replacing t with a hyperplane parameter, since the representation and the variation measure then have a different geometry.

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